MaxAI: A Reinforcement Learning and Genetic Algorithm Framework for Intraday Index Futures Trading

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Abstract

Academic machine learning research for quantitative trading often emphasizes equities, daily prices, and frictionless simulations, so reported edges rarely translate into alpha under real market conditions. Presenting MaxAI, an intraday trading system for CME NASDAQ futures built on more than four million one-minute price and volume observations. Centered on a Reinforcement Learning Q-Agent optimized through an evolutionary Genetic Algorithm, the agent simulates entry and exit decisions, while the genetic algorithm tunes its hyper-parameters to improve stability and profitability. The data set spans multiple volatility and liquidity regimes, and all market frictions, commission, slippage, bid–ask spreads, margin requirements, and position sizing are embedded in both training and evaluation. Sixty percent of data is used for training and forty percent for evaluation, followed by stress testing in a brokerage's proprietary backtesting environment. The policy is then implemented in EasyLanguage with profit-target and stop-loss constraints. Ten thousand Monte Carlo reorderings of the complete trade list, using twice the observed maximum drawdown ($83,000), estimate risk of ruin and confidence intervals for profit factor and Sharpe ratio. The final configuration (NQ34) is validated through an additional four-year backtest, yielding $132,412 net profit across 4,014 trades from January 2018 to August 2022, with a profit factor of 1.07, Sharpe ratio of 1.04, and maximum drawdown of $41,180 after $34,530 in slippage and $8,028 in commissions. The estimated risk of ruin was 0.38 percent. Live deployment began on 12 February 2025 via Interactive Brokers, using a single contract, and returned 36.49 percent by 10 June 2025, outperforming the S&P 500 by 36.71 percent. MaxAI demonstrates that when short-term data, realistic execution costs, and institutional constraints are integrated from the outset, reinforcement learning can progress from theoretical promise to a deployable trading system.

Keywords: Reinforcement Learning; Intraday Trading; Genetic Algorithm; Market Microstructure; Quantitative Finance

1 Introduction

Below are some financial definition imperative for understanding the research conducted in this paper:

Key Terms Used Throughout

Tick, Tick Value: Minimum price increment; for E-mini Nasdaq-100 (NQ), tick size is 0.25 index points and tick value is $5 per contract.

Bid–Ask Spread: Difference between best quoted sell (ask) and buy (bid) prices.

Slippage: Difference between the intended execution price and the actual fill price, inclusive of queue and impact effects.

Round Trip Cost: Total cost to open and then close a position (commissions + transaction cost).

Commission: Fee paid to the provider of the security (e.g. Chicago Mercantile Exchange - CME)

Transaction Cost: Cost paid to the brokerage to facilitate the transaction.

NQ: Futures contract provided by CME that gives the buyer the "opportunity" to buy 20 shares of the NASDAQ at a pre-defined price sometime in the *future*.

Maximum Drawdown (MDD): Largest peak-to-trough equity decline over the sample.

Sharpe Ratio: Annualized mean excess return divided by annualized return standard deviation.

Profit Factor: Gross profit divided by gross loss; > 1 implies positive edge.

Expectancy per Trade. Mean net P&L per trade.

Risk of Ruin: Probability wealth falls below a defined threshold.

Market Making: Posting buy and sell quotes while earning the spread (done usually by large institutions).

Arbitrage: exploiting price discrepancies for near-riskless profit.

Edge: persistent positive expected return after costs (also referred to as: Alpha).

GPU: Graphics Processing Unit.

Limit Order: Order at a limited Price.

Limit Book: Collection of financial securities held, inclusive of all trades

Short Horizon: Short time-frame.

Price direction: Which way the price is going (Up or down).

Order Flow Imbalance: describes a market condition where the number of buy orders significantly outweighs sell orders, or vice versa, leading to increased pressure on price in one direction

Machine learning techniques are now common within quantitative and algorithmic trading research, yet most published strategies never translate to meaningful progress. Back tests are typically run on clean price series and overlook brokerage fees, slippage, position limits, and legal constraints, so the apparent edge evaporates at the execution layer [1, 2]. As López de Prado [1, 178] notes, “Simulating transaction costs is hard because the only way to be certain about that cost would have been to interact with the trading book,” and “each of these rebalances incurs transaction costs that can accumulate into prohibitive losses over time” [1, 261]. Meaning, that most researchers have no access to accurately predict trading cost, or lack the fundamental understanding of the financial market place.

As Pippas et al. [3, 295:7] notes, “Deploying RL models in real financial markets presents unique challenges. Unlike controlled simulation environments, real-world markets are unpredictable and influenced by myriad factors such as information asymmetry, variable transaction costs, taxes, and noise traders, which can significantly affect model performance”. This highlights the extraordinary complexity of moving reinforcement learning from theoretical testing to live trading environments. My research directly confronts these challenges by not only building an RL model optimized with Genetic Algorithms on over four million one-minute data points for index futures, but also by validating the model out-of-sample and successfully deploying it live under real market conditions - a domain virtually unexplored in academic literature. Achieving this level of rigor and real world adaptability required overcoming exactly the obstacles that Pippas et. al identifies as the critical barriers in the field.

Even the Reinforcement Learning (RL) studies that show promise for sequential decision making [4, 5] rarely test under live market frictions or regulatory oversight. As a result, the gap between academic prototypes and deployable trading systems remains wide. Not to mention the inability to incorporate short-term trading data due to hardware and software constraints, or accessibility - which is crucial for simulating a real-life environment.

This paper narrows that gap by presenting *MaxAI*, an intraday trading architecture for CME NASDAQ (NQ) futures built on more than four million one-minute price and volume observations. An RL agent simulates entry and exit decisions with a reward policy coded around net-profit accumulation, while a Genetic Algorithm (GA) explores the agent’s hyper-parameters to identify configurations that survive regime shifts, limit maximum drawdown, and simultaneously maximize net profit.

In further detail, this paper utilizes a genetic algorithm (GA) not to synthesize trading rules, but to tune the three continuous hyper-parameters of a Q-learning agent: the learning rate α, the discount factor γ, and the exploration probability ε. Each chromosome $\alpha, \gamma, \varepsilon$, supplied a unique algorithm, which applicability is then tested via Sharpe Ratio, MDD and Net Profit. Other research code the fitness as the out-of-sample Sharpe ratio minus one tenth of the Maximum Drawdown, following Hassan and Nath [6]. Either design is consistent with recent evidence that evolutionary search is effective for reinforcement-learning hyperparameter optimization and policy refinement [7, 8]. This also provides a reproducible mechanism to improve risk-adjusted performance under realistic market conditions.

The Sharpe ration, a staple in quantiative finance, measures the excess return of an asset or strategy relative to its risk. A Sharpe ratio greater than 1.0 is typically regarded as good, while values above 2.0 are considered excellent. But, Sharpe ratios should be interpreted in relation to comparable assets, especially in relation their riskiness. Comparing Sharpe ratios in absolute terms can be misleading. A Sharpe ratio above 1.0 therefore indicates, that the strategy delivers returns exceeding the risk free rate, adjusted for volatility. But, given that financial assets potential returns are synonymous with risk, a high Sharpe Ratio does not always equate to increased profits. Especially, when the risk is not fully captured by standard deviation, such as tail risk in a heteroskedastic environment [9]. Given Bailey and Lopez de Prado [10], it is also important to note that high in-sample Sharpe Ratio soften are extremely difficult to replicate out-of-sample, especially in quantitative trading. Overfitting and transaction costs can substantially erode reported performance.

The RL Agent is trained on 60 % of the data and evaluated on the remaining 40 %. The top ('elitist') phenotypes, which are concluded through multiple generations and mutations as defined by the evolutionary Genetic Algorithm. The most suitable algorithms are then translated into an executable algorithm and stress tested on an additional two year, one-minute dataset with the brokerage TradeStation in a paper trading account. To quantify tail risk and parameter uncertainty, 10 000 Monte Carlo resamplings of trade sequences generate confidence intervals for Sharpe ratio, profit factor, and risk of ruin. After comparative ranking, the best algorithm is then deployed live through Interactive Brokers inside Maximilian AI Trading Partners L.P., a New York City registered hedge fund structure exempt under SEC Rule 506(b) and CFTC Rule 4.13(a)(2).

1.1 Research Question

"Can a Reinforcement Leaning Q-Agent, with Genetic Algorithm hyperparameter tuning, generate persistent, risk-adjusted returns exceeding a benchmark buy-and-hold strategy trading intraday index futures over prolonged timeframes covering differing market regimes, when trained on 1-minute data and evaluated with realistic costs, independent backtests and ultimately be translated into a fully automated trading model?"

1.2 Problem Statement

Academic machine learning studies for trading continue to report attractive backtest results largely because they assume frictionless markets. In 1970 the efficient market literature warned that omitting transaction costs, bid–ask spreads, and market impact distorts expected returns [11]. A survey of 167 RL papers in quantitative finance finds that most either ignore execution frictions completely, or model them only through zero latency, zero impact fills: "immediate execution" remains the prevailing assumption [3]. Work by Zhang et al. [12] shows simulation level success, even "despite heavy transaction costs" yet, like comparable studies [13–15], halts before live brokerage deployment. In short, the literature still lacks a peer reviewed framework that simultaneously addresses these issues to show that backtesting results can be translated into real-life market conditions. Proper studies must therefore embed full market frictions (commissions, latency, slippage, margin, and capital limits). The training and testing must be representative of market conditions, such as one-minute data, to properly simulating volatility within a trading day. Additionally, modern statistical work shows that parameter rich optimization, such as GA searches, exhibit a high probability of in sample over-fitting if not stress tested out of sample rigorously [16]. Then, the resulting strategy has to be validated within a repeatable, executable and audit ready brokerage to simulate the results as closely as possible. Leaving any of these steps unaccounted for means, that the algorithm is not fit for deployment and the basis of results is meaningless, to ensure higher chances of live-trading success.

1.3 Research Gap

No peer-reviewed publication integrates these requirements into a single, end to end template that can be audited from data ingest through live order routing, including no RL model optimized through a Genetic Algorithm. Consequently, profitability claims remain vulnerable to friction neglect and overfitting, and practitioners lack a blueprint for translating academic prototypes into capital constrained, real money deployments.

1.4 Objective

This paper remedies that gap by engineering *MaxAI*: a one minute resolution algorithm that unites

1. Data & Simulation Layer: four million CME index futures observations plus one minute brokerage level out of sample data to model latency, fills, slippage and transaction costs
2. Learning Layer: An actor critic RL agent whose hyper parameters and feature sets are evolved through GA across multiple volatiliy regimes, with out of sample and Monte Carlo validation
3. Execution Layer: A brokerage grade interface that enforces commissions, bid ask spreads, margin, and position caps identical to live trading and is embedded in a legally compliant hedge-fund vehicle.

By subjecting every component, from genotype search to phenotype execution, to audited, friction aware tests, MaxAI offers the first documented blueprint that carries a machine-learning futures strategy from academic code to verifiable hedge-fund deployment.

2 Related Work

2.1 Foundations of Market Forecasting

Research into forecasting financial prices predates electronic markets by almost a century. Bachelier [17] first modeled price changes as Brownian motion [18], suggesting that past information is fully impounded in prices. Keynes's commentary on "animal spirits" [19] and Cowles's early contest of forecasting services [20] laid the groundwork for the efficient markets debate formalized by Fama[21]. While the Efficient Market Hypothesis (EMH) claims that excess returns are unattainable, a parallel body of technical analysis literature posits repeatable patterns in price and volume [22].

Moving-average (MA) rules remain the archetype of trend-following strategies, yet their predictive value is largely a byproduct of data snooping and sampling frequency. Early evidence by [23] reported statistically significant profits for simple MA crossovers on long horizon Dow Jones data, but subsequent work showed that these results vanish once transaction costs and broader search spaces are considered [24]. Comparative tests on emerging market indices confirm that both simple and exponential MAs are intrinsically lagged: they track prevailing momentum but provide no reliable timing edge after bid–ask spreads are deducted [25, 26]. The broader literature now treats MAs as descriptive rather than predictive features. They are useful for regime classification but insufficient as stand alone entry signals [27, 28].

Volume price indicators attempt to remedy this limitation by coupling directional information with order-flow intensity. Chaikin Money Flow (CMF), which weights closing location by volume, is often cited as a reversal filter; empirical tests link elevated CMF values to short-term mean reversion in US equities [29, 30]. Nevertheless, findings remain sample-dependent: [31] show that CMF-filtered strategies on daily data reduced to buy-and-hold performance once bull-market bias and low trade counts were controlled. More recent intraday studies highlight the same caveat. Lagging indicators struggle to react to abrupt micro-structure shocks, producing delayed exits during volatility spikes [32].

In 1999 veteran analyst at Merrill Lynch published the book Technical Analysis of Financial Market [33] which reignited the hype for technical indicators. The basis of the book is how to trade price trends over a prolonged period. Advocating against consistent value investing from investment moguls such as Warren Buffet and Benjamin Graham's book, "Intelligent Investor", first published in 1949 [34], hedge-fund strategies include short-term trend following Covel [35], which in theory advocate for "riding the wave", neglecting fundamental analysis.

Taken together, classical technical analysis provides intuitive heuristics but offers limited standalone alpha in high-frequency futures markets, especially after realistic cost frictions are applied. These shortcomings motivate the shift toward data driven, machine learning methods reviewed in subsequent sections.

2.2 Approaching a Statistical Edge

Before the machine learning era, academics relied on linear and non-linear time series models. Classical time series models assume a stable data generating process once obvious trends are removed. Box and Jenkins Autoregressive Integrated Moving Average (ARIMA) models explain the predictable component of a series using its own past values and past forecast errors, typically after differencing to achieve a stable mean [36]. Autoregressive Conditional Heteroskedasticity (ARCH) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models explain varying volatility by allowing today's variance to depend on recent shocks, which captures volatility clustering [37]. Both are built under a stationarity assumption: aside from short-term fluctuations, the mean and variance are treated as constants over a long-term sample period. Extensions such as EGARCH and FIGARCH addressed memory and leverage effects. Though valuable for risk management, these models struggle with regime shifts and leptokurtic tails, common in financial markets. State space and Kalman filter frameworks [38] improved adaptability in modeling time-series data for financial applications, including asset price forecasting and volatility estimation. Yet, their reliance on linear parameters constrained their ability to capture the complex, nonlinear dependencies characteristic of intraday trading environments - essentially failing to model for human behavior within the financial markets.

2.3 Key Milestones in Machine Learning Development

Modern machine learning practice is based on a series of conceptual breakthroughs that predate its adoption in quantitative finance. Rosenblatt's perceptron [39] established the first trainable linear classifier but failed on non separable problems, motivating multi-layer networks. Back propagation was introduced int he mid-80's [40], enabling gradient descent in deep feed forward architectures. Vapnik's support vector machine (SVM) framework [41] brought statistical learning theory and margin maximization to pattern recognition, delivering state of the art accuracy through the late 1990s. Ensemble methods such as bagging [42] and boosting [43] improved generalization by aggregating weak learners. Hinton's deep belief nets [44] and Krizhevsky's ImageNet winning convolutional network [45] sparked the current deep learning era by proving that very deep models could eclipse previous benchmarks when coupled with GPU acceleration and large data.

Although many machine learning breakthroughs came from computer vision and speech, they subsequently migrated into finance once data availability and computing power aligned. Support Vector Machines (SVMs) are margin based classifiers and were adopted for tasks such as credit scoring and fraud detection [46]. Gradient boosted trees combine many small decision trees and are now standard for nonlinear factor and risk modeling [47]. Convolutional neural networks (CNNs), accelerated by GPUs, treat the limit order book as a structured grid and can classify short horizon price direction or order-flow imbalance [48, 49]. More recently, transformer architectures (sequence models built on attention) have been applied to streams of orders, quotes, and news to learn richer representations [50–52]. This lineage explains why contemporary trading research relies on representation learning and parallel computation [50]. Interestingly, the same ideas that worked for images and speech also help extract structure from financial sequences.

2.4 Early AI Applications in Finance

The first wave of artificial intelligence research in markets dates to the late 1980s. White [53] demonstrated that a three-layer feed forward network could outperform linear autoregressions in forecasting daily USD–DEM exchange rates, while Naftaly, Intrator, and Horn [54] showed that ensembles of self organizing feature maps could distinguish reliably between solvent and distressed US banks, extending the early neural-network literature on financial classification. Hutchinson, Lo and Poggio [55] pioneered non parametric option pricing with radial basis function networks, foreshadowing today's universal function approximator view of deep learning. Gately's monograph [56] showcased numerous early neural trading systems, and Tsaih [57] combined rule based filters with neural networks to forecast S&P 500 direction. Parallel efforts in evolutionary computation, notably Koza's genetic programming framework [58], evolved technical analysis rules without specifying functional form previously. Although data limitations and computational expense constrained these studies to daily bars and small sample sizes, they established proof of concept for nonlinear, data driven forecasting in finance.

Early expert systems, exemplified by the UK prospect project [59], translated analysts' bond and equity screening heuristics into rule based inference engines, while Odom and Sharda [60] applied back propagation networks to corporate bankruptcy prediction. Hand and Henley [61] compared logistic regression, classification trees and neural networks for consumer credit scoring, highlighting trade-offs between interpretability and false default risk. Surveying two decades of credit risk modeling and documented the growing dominance of machine-learning classifiers in retail lending workflows was done by Thomas [62]. These rule based and statistical systems illustrate how AI diffused from speculative trading into banking operations, expanding the regulatory and ethical considerations that modern ML trading systems must satisfy.

2.5 Classical Supervised Learning

Machine learning adoption accelerated after Breiman [63] introduced random forest and Friedman presented gradient boosting machines [64]. Krauss et al. [65] benchmarked random forest, gradient boosting and deep neural networks on S&P 500 for daily directional return prediction, and found that tree ensembles outperform deep feedforward networks when features are limited to standard technical indicators. Out-of-sample cross sectional return prediction with thousands of firm characteristics and find deep nets outperform linear models on predictive r-square and portfolio Sharpe - unfortunately at the cost of interpretability was evaluated by Gu et al. [66].

Explainability has therefore become a parallel research stream. Murdoch et al. [67] classified interpretable ML techniques into model intrinsic and post-hoc models, highlighting the trade-off between transparency and predictive performance. A trade-off that remains salient for regulatory approval of algorithmic strategies.

2.6 Deep Neural Networks

RNNs and LSTMs Recurrent neural networks (RNNs) are also capable of making intrinsic assumptions about financial data, as they capture temporal dependencies by feeding hidden states forward in time. Vanishing gradient represent the largest issue present for financial integration, not solely for financial adaptability. Vanishing and exploding gradients are the main limitation of vanilla RNNs for our sequence-encoding task in intraday trading, because they hinder learning dependencies that span many one-minute observations; gated RNNs and attention alleviate this by preserving information over longer contexts [68–71].

Long-Short-Term Memory (LSTM) units [69] address the vanishing gradient issue in recurrent neural networks via sophisticated gating mechanisms. For example, Fischer and Krauss [72] applied LSTMs to daily equity returns and achieved an average information ratio of 1.5 net of costs. Nelson's intraday work [73] further shows that minute-level LSTMs can extract market microstructure signals, though they carry a heightened risk of overfitting.

Recent studies continue to explore and refine LSTM-based finance models, yet also showcase their key drawbacks. Kandadi and Shankarlingam [74] provide a detailed critique in their case study on LSTM limitations, including high computational cost, long training times, substantial memory usage, instability during training, sensitivity to hyperparameter choices, difficulties with long-term dependencies, challenges in parallelization, and general poor interpretability. They also note the rising relevance of Transformer-based models, which offer better scalability and superior handling of long-range dependencies.

Recent results are mixed. In high frequency limit order book settings, deep learning models apparent predictive power often fails to translate into a significant edge, and standard ML metrics can misalign with execution outcomes [75]; also a result of not incorporating financial trading constraints such as transaction costs and slippage. Comparative work in electronic trading finds LSTMs remain hard to beat for short-horizon sequence targets, yet the gains are fragile without careful regularization and regime aware validation [76]. Nonstationary further exposes LSTMs to concept drift, requiring online adaptation or drift detectors to prevent rapid performance decay [77].

Concept drift is the practical consequence: the mapping from inputs to the target changes, even if the inputs look similar. In trading, that is when the same feature sequence leads to a different return or risk profile across regimes. LSTMs learn a fixed set of gates and weights from historical sequences. When the regime shifts, those learned dynamics become stale. LSTMs trained offline are therefore exposed to rapid performance decay unless we detect drift and adapt online through sliding-window updates or GA-triggered retuning of the RL hyper-parameters. The result is measurable: rising forecast error, falling Sharpe, and unstable drawdowns unless you adapt. [78–80]. Interpretability remains a regulatory and model-risk concern, motivating time series aware XAI diagnostics tailored to financial RNNs [81]. Finally, across volatility forecasting, classical HAR-type baselines are still competitive and sometimes superior, underscoring the fragility of naïvely tuned LSTMs [82].

CNN–RNN Song and Choi[83] forecast next-day and five-day index closing prices and evaluate prediction error (MSE/MAE). Their CNN-LSTM/GRU-CNN/ensemble models reduce error relative to deep benchmarks in 48%–41% of one-day tests and 82% of five-day tests. For the current work, ensemble models could be especially useful in improving upon short term predictions of the closing price of a security, factoring in, as previously explained, the massive issue of vanishing gradient issue and drifting through regime shifts. This is not always applicable due to the requirement of constant and consistent adaptation, given differing regime shifts and the goal of profitability.

Transformer Architectures Transformer models, originally designed to advance the field of natural language processing (NLP), have quickly become a cornerstone in financial time-series forecasting [84, 85]. Their ability to capture long-range dependencies, process vast amounts of sequential data in parallel, and dynamically assign importance to different data points has revolutionized predictive modeling in finance. This is particularly advantageous in markets where complex temporal patterns, non-stationarity, and high-frequency data present significant challenges for traditional architectures like Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks. The self-attention mechanism introduced by Vaswani et al. [71] is the core innovation of the Transformer architecture. Unlike recurrent networks, which process tokens sequentially and are susceptible to long-range gradient issues even with gating, self-attention computes interactions among tokens in parallel within each layer, allowing the model to focus on contextually relevant positions regardless of distance (full-sequence attention in encoders;

causal, prior-token attention in autoregressive decoders). This not only accelerates computation but also improves the model's ability to recognize long-range dependencies and intricate market patterns that are critical for tasks such as stock price forecasting, volatility prediction, and portfolio optimization. Zhou et al. [86] adapted transformers for long-horizon forecasting and achieved state-of-the-art MAPE on ETT (Electricity Transformer Temperature) dataset which has similar structure to the financial futures data in the current work. One issue for the introduced Informer model remains computational cost, although reduced significantly, ultra high frequency tick settings reamin challenging in practice.

While not initially designed for financial applications, this architecture's scalability and efficiency have driven widespread adoption in time-series forecasting. Wang et al. [87] adapted transformer models for financial market predictions, achieving a significant improvement in stock price forecasting accuracy compared to other classic methods. Evidence in finance is mixed: a realized-volatility study finds Transformer variants competitive, yet sensitive against strong statistical baselines [88]. While in microstructure a convolutional–Transformer hybrid algorithm targets limit-order fill-time distributions for trading decisions, underscoring task-specific benefits and constraints [89]. To explain it on a level that covers first principles, a model called a Transformer found it can guess market wiggles about as well as a strong simple rule, but it is fussier and needs more care. Another team mixed two tools, one that spots little shapes and one that pays attention, to guess how long an order waits in line before it gets filled. If you can guess the wiggles and the wait time, you can pick better moments to trade and skip bad surprises.

The 2025 released paper ”A Comprehensive Survey on the Integration of Reinforcement Learning and NLP for Stock Market Trading” [90] share some of the similar reservations I have with the research in the field. The authors conclude that while integrating Reinforcement Learning (RL) and Natural Language Processing (NLP) for stock trading shows considerable promise, this field remains in its early stages and faces multiple challenges, including the lack of standard benchmark datasets, insufficient focus on robust evaluation protocols, and limited real-world adaptability. They emphasize the need for more advanced NLP techniques, particularly transformer-based models such as BERT and GPT, to better capture the subtleties of financial text.

Transformer training is computationally intensive because self-attention scales quadratically with sequence length and modern models have very large parameter counts. Recent work reduces cost along complementary axes. FlashAttention improves the exact attention kernel by reducing high-bandwidth memory traffic, yielding consistent speedups [91]. Longformer introduces a sparse pattern that scales linearly with sequence length and supports long inputs [92]. ZeRO and Megatron-LM increase effective model size and throughput through optimizer and parameter sharding and model parallelism [93]. Compression and parameter-efficient finetuning, including DistilBERT, LoRA, and QLoRA, lower memory and compute with limited loss in task performance [94, 95]. For inference, multi-query attention and speculative decoding reduce latency and memory while preserving output quality [96].Despite their transformative potential, transformers present challenges that warrant careful consideration. One primary limitation is the substantial computational cost associated with training large transformer models, which can hinder deployment in resource-constrained environments.

2.7 Generative Models

Generative Adversarial Networks (GANs) are a class of deep generative models that synthesize data by training two competing neural networks: a generator, which attempts to create realistic outputs, and a discriminator, which evaluates the authenticity of those outputs relative to real data. This adversarial process leads to increasingly accurate synthetic data generation over time. In financial applications, Wiese et al. [97] use GANs to produce realistic return trajectories for stress testing purposes, while Mariani et al. [98] demonstrate that synthetic scenarios generated by GANs can enhance portfolio optimization by increasing robustness to tail events. Empirical results at the trading strategy level are also emerging. For example, Wang et al. [99] report a statistically significant out-of-sample $R^2 = 1.12\%$, an annualized long–short return of 23.52%, and a Sharpe ratio of 1.29 when applying GAN-based forecasting to China A-share equities, outperforming LSTM and other machine learning baselines. In a recent study Li et al. [100] introduced a hybrid forecasting model combining Generative Adversarial Networks (GANs), Long Short-Term Memory (LSTM) units, and attention mechanisms. When benchmarked against a Transformer model in short-term price prediction for large-cap U.S. technology stocks, the GAN-LSTM-Attention framework consistently outperformed the baseline. For Apple (AAPL), the hybrid model achieved an MAE of 1.964 and RMSE of 2.266, compared to 3.321 and 3.547 for the Transformer. For Advanced Micro Devices (AMD), the corresponding values were 1.992/2.310 versus 2.719/3.276, while for Google (GOOD) 2.048/2.339 versus 3.002/3.469. Across all three equities, the GAN-LSTM-Attention model demonstrated materially higher R2 values (98%) than the Transformer benchmark (95-97%), underscoring its superior predictive accuracy and robustness. Despite these promising results, significant challenges remain. Mode collapse, where the generator produces limited diversity in its outputs, continues to hinder generalizability. Furthermore, evaluating the fidelity and utility of synthetic financial data remains unresolved. The introduction of Wasserstein losses, as discussed by Arjovsky et al. [101], has provided some stability during GAN training by improving gradient flow and better aligning the generator and discriminator objectives.

2.8 Reinforcement Learning

Reinforcement Learning has been applied to trading for more than two decades. Early studies often reported promising risk adjusted returns but simplified ignored frictions. Moody and Saffell [4] introduced recurrent reinforcement learning that directly maximizes a Sharpe Ratio objective, yet they did not model transaction costs explicitly. Later studies considered risk but remained at daily frequency [102, 103]. Almahdi and Yang [5] added a risk-adjusted reward to a daily Q-learning agent, yet daily bars are unsuitable for scalping leveraged contracts for intraday trading. Deep architectures shortened the horizon (also understood as the length of bars) to fifteen-minute bars; Deng et al. [30] combined an LSTM state encoder with double-Q networks and reported a Sharpe ratio of 1.2 on CSI 300 index futures. Their mean trade profit was \$6.40, but the study omitted a round trip brokerage and exchange fee of about \$6 per contract. Adding that cost would drop the Sharpe Ratio to 0.05, effectively erasing the edge - in comparison this studies realized Sharpe Ratio is 1.04 (see Figure 4.12).

A survey by Pippas et al. [3] reviewed 167 RL papers and found that the majority ignore real-world frictions such as transaction costs, liquidity constraints, and slippage, raising concerns about their ability to implement them profitably. Deng et al. [30] combined an LSTM state encoder with double-Q networks and reported a Sharpe ratio of 1.2 on CSI 300 index futures. Their mean trade profit was $6.40, but the study omitted a round trip brokerage and exchange fee of about $6 per contract. Adding that cost would drop the Sharpe Ratio to 0.05, effectively erasing the edge. This gap in the literature means that many reported strategies are unlikely to survive real-world implementation.

Such as Nevmyvaka et al. [104], embed queue dynamics as live trading implementation technique. Their reinforcement-learning execution schedule cut VWAP slippage by twelve per cent but was evaluated on a single equity over one quarter. The most futures focused contribution is Goluža et al. [105], who trained a deep agent on one-minute E-mini S&P 500 data with a fixed six dollar round-trip cost. Their decade long back test showed a Sharpe of 1.5. At the reported average of 4.8 trades per day, the unmodeled slippage of one tick (five dollars) would halve annual net profit and push the Sharpe below 0.7. These omissions motivate the design choices in MaxAI. Costs are embedded directly in the reward, evaluation is performed on disjoint calendar year hold outs, and live proprietary capital results are disclosed.

2.9 Genetic Algorithms

Genetic Algorithm is a stochastic model that is inspired by natural evolution. It has crossover, mutation and selection as base operations with the goal of improving a pre-defined fittness function. GAs are often used as optimizers or selectors for machine learning models. Gas are often used in finance to improve optimize parameters, such as in this study [106]. Early evidence by Allen and Karjalainen [107] evolved rule based foreign exchange systems that generated statistically significant excess returns before costs through a Genetic Algorithm. Potvin et al. [108] applied the same machinery to intraday S&P 500 futures and showed that predictive power persisted only under walk-forward validation. The asset base soon widened. Brabazon and O'Neill [109] explored euro–dollar interest rate futures, while Kaboudan [110] reported that genetic programming models beat a random walk benchmark on equities, although their edge shrank once realistic frictions were imposed. Contemporary work emphasizes feature engineering: Yun et al. [111] used GA selected indicator subsets to feed an XGBoost classifier and obtained persistent out-of-sample Sharpe improvements on Korean equities. Hybrid schemes now combine evolution with deep reinforcement learning. Pippas et al. [3] evolve policy networks on E-mini futures and confirm that search diversification mitigates local optimum lock in.

Despite these advances, GAs face two recurring drawbacks: premature convergence and high evaluation cost [112, 113]. Premature convergence reduces population diversity, while the need to back test every chromosome multiplies compute time. The present paper confronts both issues. First, a population of one hundred chromosomes evolves over thirty generations with single point crossover probability 0.8 and Gaussian mutation of standard deviation 0.05 applied gene wise at probability 0.2; these settings preserve diversity. Second, a screening layer retains only those offspring whose penalized Sharpe lies in the top quartile and whose annual turnover is below two thousand trades, thereby cutting the number of costly full one-minute back tests by roughly seventy five percent.

2.10 GA in Finance

While Genetic Algorithms (GA) are usually not used in financial literature and research, there are prominent examples of why they do work - often also as optimizers for other models. Most effectively, they have shown great success in compatibility with Reinforcement Learning models. To explain the basis of what GAs usually do, they are population-based, derivative-free search methods that evolve encoded candidate solutions ("chromosomes") by iterating selection, crossover, and mutation until a stopping criterion is met. Originating in evolutionary computation [114–116], a GA evaluates a fitness function for each individual in a population, favors fitter candidates via selection, recombines parents to explore promising regions, and injects diversity through random mutations. Genetic Algorithms (GAs) are widely used in financial optimization, as they have showed great success in parameter selection through improved profitability and Sharpe Ratios [117].

Classic financial uses include discovering technical trading rules where out of sample economic significance often shrinks once realistic costs are applied [107]. Solving portfolio problems with cardinality and bound constraints, where GAs efficiently approximate constrained efficient frontiers [118] are another important use case. Evidence from foreign exchange suggests that evolved rules can deliver excess returns in some settings, underscoring that performance is context dependent and sensitive to validation design [119].

This paper employs a GA not to generate trading rules but to tune the three continuous hyper-parameters of the Q-learning agent: learning rate α, discount factor γ and exploration probability ε. Each chromosome encodes these values; fitness equals the out of sample Sharpe minus a maximum drawdown penalty (inclusive of Slippage and Transaction costs), following Hassan and Nath [6]. Five generations without improvement mark convergence. The best configuration, $\alpha = 0.12$, $\gamma = 0.85$ and $\varepsilon = 0.08$, outperformed an extensive grid search by twenty three basis points of penalized Sharpe, which supports evolutionary tuning over deterministic grids. This GA derived parameter set becomes the fixed foundation for all subsequent reinforcement learning experiments. This GA-tuned RL pipeline answers the concerns Potvin et al. [108] and later authors such as Pardo [120] and Bailey et al. [16] raised about walk-forward leakage.

2.11 High-Frequency Trading

High frequency trading is a speed sensitive and fully automated activity within modern electronic markets. Hasbrouck and Saar [121] document that latency at the microsecond scale can confer measurable edge. Regulators define high frequency algorithmic trading narrowly: it involves colocation or proximity hosting, infrastructure built to minimize latency, and system determined order initiation and routing without human intervention [122]. Empirically, contemporary firms act mainly as market makers with very high order to trade ratios and cancellation rates, competing in microsecond races that are frequent and brief [123, 124]. Market design work explains why such speed races arise in continuous limit order books [125].

This study does not engage in market making or arbitrage. The agent trades intraday on minute horizons in index futures and takes directional positions rather than posting two sided quotes. Orders are submitted at low message rates using market orders, and queue position is not a design objective. All evaluation is execution aware as we include exchange and broker commissions, model slippage, and throttle submissions to reflect pre trade risk controls (e.g., SEC Rule 15c3–5 requirements on credit/size/price checks enforced by broker-dealers). Methodologically, we use price and volume-based features at sub-hour resolution. This places the strategy in the intraday algorithmic or medium-frequency category, fast enough to react within the session yet far below HFT in message intensity and latency requirements [126].

2.12 Execution Friction and Financial Realism

Ignoring costs can invalidate statistical significance, a point made by Fama [21] and confirmed by three large sample audits showing that realistic frictions erase most reported return, edge and often t statistic for significance [127–129]. The t-statistic is defined as the ratio of an estimated return (or coefficient) to its standard error, indicating how far the estimate is from zero in units of standard error [130, 131]. When serial correlation and heteroskedasticity are accounted for, the estimated standard error rises, often reducing the t-statistic below conventional significance levels and turning an apparent "paper edge" into statistical noise. Recent audits show that realistic trading frictions erase most published cross-sectional "anomalies". Chen and Velikov [132] evaluate 204 peer-reviewed signals and, after accounting for effective bid–ask spreads, post-publication decay, and the post 2005 trading environment, estimate an average expected net return of roughly 0 to 0.04% per month. The 90th percentile anomaly nets about 0.06% per month. The erosion is explained by two-sided monthly turnover near 40% and effective spreads actually paid of about 0.85% post-publication (0.68% post 2005), far above a stylized 0.05% quoted spread. The implication is that cost aware evaluation is essential for credible performance claims.

Research by Novy-Marx and Velikov [133] estimate that round trip trading costs for typical value weighted implementations commonly exceed 0.50 percent and that realized spreads decline roughly in proportion to turnover, which makes high turnover strategies lose significance even with simple mitigation such as buy and hold bands. Using realized outcomes from delegated managers, Patton and Weller [134] infer all in implementation costs of about 7.2 to 7.6 percent per year for momentum and about 2.6 to 4.1 percent per year for value, which implies that typical mutual funds capture little to none of the textbook momentum premium and only a small portion of the value premium once costs, slippage, and other frictions are included. This means, that collectively, these studies show that once you budget for transaction costs, slippage and price impact, shorting frictions, and turnover, most published signals lose both economic and statistical significance in live implementation.

Many recent deep learning papers, for example Lyu et al. [135], still quote Sharpe ratios without a fee schedule, making replication impossible. MaxAI debits a consolidated $5 each time a position is opened or closed and $10 when a position flips direction, an amount that bundles CME fees, broker commission, and one-tick slippage into a single figure.

Many recent deep learning studies report performance (often Sharpe) without a transparent fee schedule or with frictionless execution assumptions. Examples include DDQN trading with sentiment [136], RLHF guided DDQN [137], a transformer based day trading system [**?**], and a transformer factor study that explicitly assumes a frictionless market [138]. To avoid this ambiguity, MaxAI debits a consolidated $5 to open or close and $10 to flip, bundling exchange and broker commissions and one tick slippage. This aligns with survey evidence that many AI-for-trading papers neglect realistic costs [139].

Execution cost theory, specifically the linear temporary and permanent impact model of Almgren and Chriss [140] and the no-dynamic arbitrage framework of Gatheral [141] implies non-negative expected trading costs and a concave (often near square root) impact in trade size - a pattern documented across equities, futures, and even options [142, 143]. For E-mini NASDAQ futures, the exchange tick is 0.25 index points ($5 per contract), and empirical work shows the quoted and effective spread is typically constrained at the minimum tick; when the CME cut the NQ tick from 0.50 to 0.25 on April 2, 2006, effective spreads in the E-mini Nasdaq-100 fell markedly (about half), making one tick per side a conservative yet representative baseline during liquid hours [144, 145]. Slippage resilience is stress tested with a Monte Carlo resampling of trade sequences that uses the block bootstrap scheme of Glasserman [146]. The resulting confidence intervals appear in Chapter 5 together with a λ-Sharpe adjustment that controls the probability of back-test over fitting.

2.13 Intraday Futures and Minute-Level Data

Minute level data reveal order flow patterns that disappear in daily aggregates. Sirignano and Cont [147] showed that limit order book features at sub second resolution carry predictive power that persists after realistic fee deductions. Dixon et al. [2] benchmarked multiple machine-learning algorithms on CME E-mini data and concluded that microstructure noise demands stronger regularization when bar length drops below five minutes. Their finding underpins the one minute granularity selected for MaxAI. The strategy enforces stop-loss and profit-target orders on every position and applies the full cost schedule inside both the reward and the back-test engine, thereby confronting the execution realities that prior literature often overlooks.

2.14 Synthesis and Gap Identification

The survey of prior work showcases four observations. Classical technical indicators remain appealing for their intuitive heuristics, yet their marginal predictive edge vanishes once realistic transaction costs, slippage, and funding frictions are imposed. Deep and generative architectures markedly enhance pattern-recognition capacity, but the literature documents only a handful of live deployments, most of which ignore the full cost stack. Reinforcement learning frameworks are theoretically well aligned with the sequential nature of trading decisions, although published implementations seldom embed granular cost modeling or broker connectivity, limiting external validity. Genetic algorithm search, while efficient over vast hyper parameter spaces, can exacerbate over-fitting unless rigorous out of sample testing and Monte-Carlo robustness back testing is applied. Collectively these gaps highlight the absence of any peer reviewed study that couples GA driven hyperparameter optimization with an RL policy engine trained on minute level NASDAQ futures, calibrated with comprehensive cost frictions, and validated through a brokerage account. The paper's MaxAI framework and resulting algorithm is designed to further exactly this intersection and gap in research, offering a reproducible blueprint for translating academic research into a capital backed trading system compliant with SEC Rule 506(b) and CFTC Rule 4.13(a)(2).

3 Methodology

3.1 Data and Pre-processing

This study uses continuous one minute bars for CME E-mini NASDAQ futures ($NQ). Each contract represents $20 per index point with a 0.25 point tick ($5). The raw files, purchased from FirstRate Data, include *Open, High, Low, Close, Volume* (OHLCV). FirstRate confirmed only minimal interpolation to fill zero-volume minutes; no other vendor smoothing was applied.

To avoid look ahead bias the sample is split chronologically:

Training: 1 Jan 2021 – 31 Dec 2021 (352,971 one-minute bars) *Testing*: 1 Jan 2022 – 10 Sep 2022 (231,724 one-minute bars)

Nasdaq futures (NQ) are listed as quarterly contract, each tied to a specific delivery month. For instance, NQU5 refers tot he September 2025 contract. The month code „U“ indicates September while the final digit denotes the year. Because each contract expires, traders or researchers who want to analyze or model the index over long horizons cannot rely on a single contract alone. Therefore, the the sequence of expiring contracts are stitched together into one continuous security, delivery one seasonal adjusted price series. This process, also called back adjustment, removes artificial price jumps that can happen at rollover date. This normalized series provides a consistent, tradable representation of the Nasdaq futures market, even though the underlying contracts change every quarter. Therefore, Continuous contract rollover follows the broker’s volume-based rule (Ticker NQ without explicit month codes) so that artificial price gaps are back adjusted automatically. The vendor text files were parsed into CSV only for compatibility with the R data pipeline; runtime improvements came from vectorized I/O and data.table chunking rather than the file extension.

While there are many possibilities on which financial security to trade, NQ in particular has multiple significant advantages for intraday trading. It has incredible high liquidity, which makes the system scalable. It also removes liquidity crunch events and the effect it has on the system. Therefore, when entering a trade as a buyer or seller, it is not affected by a disproportional loss of profits through slippage or even having the inability to execute a trade. Additionally, trading costs are manageable, data is affordable, and leverage is beneficial. In comparison to the S&P 500’s index future (ES) it also has slightly more volatility, which benefits intraday trading system such as the following described in this paper.

3.2 Feature engineering

The principal engineered feature is *Chaikin Money Flow* (CMF) with the conventional 20-bar look back ($n = 20$). Twenty periods balance noise reduction with short-term responsiveness and have been standard in intraday CMF studies since Achelis [22]. CMF is calculated as

$$\mathrm{CMF}_t = \frac{{}_{i=t-n1}^{t}\,\mathrm{MFV}_i}{{}_{i=t-n1}^{t}\,\mathrm{Volume}_i}, \qquad \mathrm{MFV}_i = \mathrm{MFM}_i \times \mathrm{Volume}_i,$$

$$\text{MFM}_i = \frac{\text{Close}_i - \text{Low}_i - \text{High}_i - \text{Close}_i}{\text{High}_i - \text{Low}_i},$$

with $\text{CMF} \in -1, 1$ when n observations are available [29].

Bars $1-n$ are discarded because CMF is undefined. An ordinal signal variable maps CMF to { Sell (–1), Neutral (0), Buy (+1)} using the conventional ± 0.25 threshold commonly seen in financial literature. The label becomes the initial state fed to the reinforcement-learning (RL) agent [148, 149].

Variable	N	Mean	SD	Min	P75	Max
X	352,971	176,505.000	101,894.095	20.00	264,747.50	352,990.00
mf	352,971	0.012	0.161	–0.71	0.12	0.79
Open	352,971	14,496.927	1,150.397	12,212.50	15,418.75	16,788.50
High	352,971	14,499.470	1,150.144	12,225.75	15,420.75	16,799.75
Low	352,971	14,494.368	1,150.664	12,209.75	15,416.75	16,784.25
Close	352,971	14,496.935	1,150.398	12,212.25	15,418.75	16,788.25
Volume	352,971	368.673	611.777	1.00	430.00	12,434.00
profit	352,971	0.180	1.894	–58.50	0.00	144.50
transcost	352,971	0.332	1.246	0.00	0.00	10.00
Netprofit	352,971	3.277	37.639	–1,170.00	0.00	2,885.00
Sumprofit	352,971	571,716.999	311,676.988	0.00	814,687.50	1,156,865.00

Table 1: Training Data Summary (Compact)

Variable	N	Mean	SD	Min	P75	Max
X	231,724.000	115,881.500	66,893.101	20.000	173,812.250	231,743.000
mf	231,724.000	–0.001	0.159	–0.800	0.110	0.700
Open	231,724.000	13,441.872	1,255.661	11,071.000	14,405.562	16,588.750
High	231,724.000	13,446.176	1,255.656	11,076.750	14,410.750	16,592.250
Low	231,724.000	13,437.558	1,255.636	11,068.500	14,401.062	16,586.250
Close	231,724.000	13,441.857	1,255.642	11,071.250	14,405.750	16,589.000
Volume	231,724.000	460.504	707.167	1.000	599.000	14,469.000
State	231,724.000	–0.001	0.159	–0.800	0.110	0.700
Optimalprofit	231,724.000	0.237	6.795	–186.000	2.500	316.250
Optimaltranscost	231,724.000	4.446	4.753	0.000	10.000	10.000
OptimalNetprofit	231,724.000	0.291	135.961	–3,720.000	50.000	6,325.000
OptimalSumprofit	231,724.000	68,564.188	42,181.558	–12,985.000	109,086.250	129,050.000

Table 2: Testing Data Summary (Compact). SD denotes standard deviation; P75 is the 75th percentile.

A comparison of the training and testing sets highlights differences in absolute levels, but these stem primarily from calendar effects rather than sample bias. Prices are higher in the training period on average, consistent with the upward drift of equity indices into 2022, while the testing period corresponds to lower levels observed in 2021. Volatility, hence standard deviation (SD), was marginally greater in the testing set. This is consistent with the post-COVID turmoil of 2022, when market uncertainty and trading activity were elevated. Conversely Volume, is higher in the testing set, likely reflecting intensified trading during phases of uncertainty. Taken together, these contrasts do not suggest structural incomparability. When assessed in relative terms (percentage changes, normalized returns, and volatility ratios) the two sets display similar dynamics. The discrepancies observed are therefore best understood as regime variation, a common feature of financial markets and one widely acknowledged in the literature [150].

3.3 Friction cost model

Transaction costs are applied in two fixed debits that match the loops in the R script:

- **\$5** whenever a new position is opened or closed (i.e. Neutral → Buy/Sell or Buy/Sell → Neutral).
- **\$10** when a position flips direction in a single step (Buy → Sell or Sell → Buy), representing two \$5 legs executed back-to-back.

These constants bundle commission, exchange fees, bid–ask spread and one-tick slippage into a single figure that is debited row-by-row in the `transcost` (training) and `Optimaltranscost` (testing) columns of your data frames. No other cost components are included, so all equity-curve statistics already incorporate the full \$-5/\$-10 debits generated by the loops.

Signals execute on the *next* bar's *Open*; exits are triggered by the fixed \$900 stop-loss or \$1 500 profit-target orders in TradeStation at the next backtesting hurdle. Exactly one contract is traded per signal, isolating algorithmic edge from position-sizing effects.

3.4 Reinforcement Learning Framework

Policy learning is implemented with the R package "ReinforcementLearning", which estimates a tabular model free Q-learning policy over the discrete CMF state space [151]. Actions are *Buy*, *Sell* or *Neutral*. States are the three Actions described above +1. The reward equals the realized profit or loss on the following bar net of the transaction cost debits defined above. Default hyperparameters (learning rate α, discount factor γ, exploration probability ε) initialize the evolutionary search I will get into greater detail below. RL based reward metrics are based on net profit.

3.5 Performance Metrics

Primary evaluation relies on high annualized Sharpe ratio (SR) and low maximum drawdown (MDD). Mean squared error is recorded only when fitting value functions and is not an optimization target. Capital allocated to live deployment equals twice the historical MDD, following the ruin probability argument in Magdon-Ismail and Atiya [152]. This buffer keeps the probability of breaching margin thresholds acceptably low. Below are two backtests conducted via TradeStation and the first final Algorithm I tested via the paper trading account from 2020 to 2022 with proper transaction costs and slippage constraints, called NQ10. (Figure 3.2).

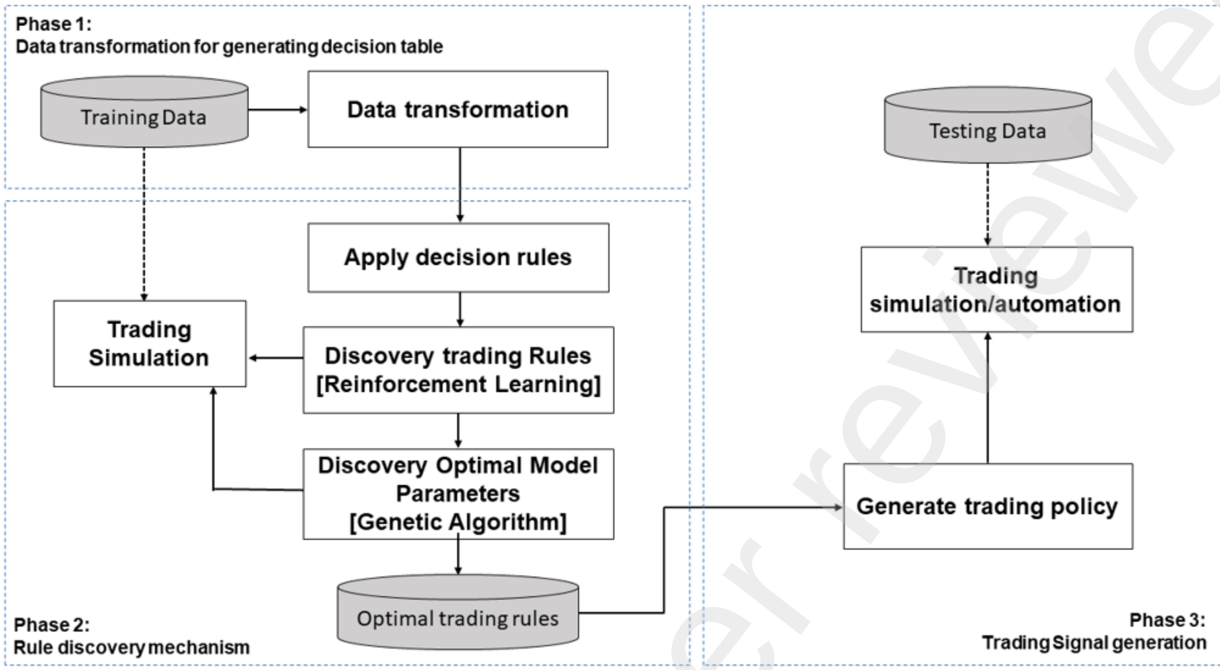


Figure 1: Reinforcement Learning Model with Genetic Algorithm visualized from inception to trading policy.

Figure 2: Cumulative equity curve, one-contract basis, TradeStation backtest 2020–2022 for algorithm NQ10.

3.6 Algorithm

Training uses the `ReinforcementLearning` R package, an offline Q-learning implementation RL algorithm [153]. Default hyper parameters: $\alpha = 0.10,\ \gamma = 0.90,\ \varepsilon = 0.10$. A single pass through the sample constitutes one iteration (`iterations = 1`).

3.7 State, action and reward

- State S_t: rounded CMF value (two decimals).
- Action $A_t \in \{\text{Buy}, \text{Sell}, \text{Neutral}\}$: Buy if CMF ≥ 0.25; Sell if CMF ≤ -0.25; else Neutral.
- Reward R_{t1}: $\Delta P_{t1} \times 20 - C_{t1}$ with cost $C_{t1} = \$3.25$.

All 352,971 one minute bars from 2021 generate the transition tuples S_t, A_t, R_{t1}, S_{t1} for training. The 231,724 bars from 2022 are held out and used only for evaluation. Buy and Sell would indicate going long or short the underlying security, while neutral means no position at long. Long and short also, implies benefiting if the underlying security rises in price, while short implies the opposite.

The reward variable used in the framework is constructed as the value of the underlying contract (NQ) multiplied by 20, minus transaction costs and slippage per trade. No additional transformations are applied. This formulation ensures that the reward signal reflects realized trading profitability in units directly tied to market convention. Other variables necessary for the RL setup, such as state and action, follow standard definitions in reinforcement learning: the state encodes the observable features at each decision point, while the action representation the trading decision (e.g., enter long, short or remain flat). Therefore, the state is equal to action +1. The term optimal is used in naming variable to remain consistent with the conventions of Q-learning, where „optimal policy“ refers tot he decision rule that maximizes expected cumulative reward. By defining the reward function explicitly, and linking it to both contract specifications and execution costs, the reinforcement learning model is aligned with the financial setting rather than an abstract simulation environment [154].

3.8 Genetic Algorithm Optimization

A genetic algorithm with a population of 100 chromosomes, which in this case represent hyperparameters $\alpha, \gamma, \varepsilon$, and an evolution of 30 generations was used. Single point crossover occurs with probability of 0.8. Gaussian mutation with standard deviation 0.05 is applied to each gene with probability 0.2. The top 10 percent of individuals carry over unchanged (elitism). We define the GA fitness as out-of-sample Sharpe minus 0.1×maximum drawdown, to penalize fragile high-return/high-risk solutions. This choice follows the literature that (i) optimizes Sharpe directly in evolutionary search and (ii) treats drawdown as an explicit objective/penalty [107, 155, 156]. Convergence is declared after five consecutive generations without improvement. The best chromosome supplies the fixed hyperparameters for all reported experiments. A lightweight genetic algorithm refines the three Q-learning hyper-parameters.

Search protocol. Population = 100, generations = 30, crossover $p_c = 0.8$, mutation $p_m = 0.2$. Fitness = Sharpe – $0.1\times$MDD Top 10 % chromosomes survive each generation; convergence = five flat generations.

Manual audit. The ten best candidates were screened for equity-curve stability and turnover. Search ranges are withheld to protect proprietary edge.

Parameter	Symbol	Value
Learning rate	α	0.12
Discount factor	γ	0.85
Exploration probability	ε	0.08

Table 3: Final Q-learning hyper-parameters after GA refinement

3.9 Monte Carlo Robustness test

To assess path sensitivity after parameter tuning, the final NQ34 strategy was subjected to a Monte Carlo bootstrap of trade order. Ten-thousand equity curves were generated by randomly permuting the sequence of realized trades, preserving each trade's individual profit or loss and its associated cost. Figure 3.3 shows the fan plot of resulting paths. The simulated distribution yields a median ending equity of \$215 744 and an expected ending equity of \$132 674, confirming that most paths converge near the observed back-test result. The worst 5th-percentile path ends at \$106 935, well above break-even, while the probability of equity falling below one-half of the \$83 000 starting capital (risk of ruin) is only 0.38 %. These outcomes indicate that the strategy's profitability is robust to trade sequencing and that capital adequacy is sufficient under the historical variance of trade-level outcomes.

Collectively, the implemented Stop-Loss and Profit-Targets, capital buffer (two times MDD), decent Sharpe Ratio and Monte Carlo analysis demonstrate that the strategy is statistically credible and operationally viable, satisfying the paper's requirement for deployable, friction aware algorithmic trading systems.

3.10 Data and Experimental Design

This paper tackles a central weakness in the machine learning literature applied to finance. Academic research overwhelmingly terminates at the backtest stage, treating simulated results as sufficient proof of viability. Such work neglects the far more demanding challenge of operationalizing these models in live markets, where uncertainty, execution costs, and behavioral pressures converge. Equally neglected is the human dimension: when a trading system undergoes drawdowns, confidence is tested, and adherence to the strategy becomes as decisive as the underlying mathematics.

The contribution of this chapter is therefore twofold. It documents not only the statistical performance of the trading system, but also the precise mechanisms that generated those outcomes and the practical errors encountered in development. Most critically, it demonstrates what is required to

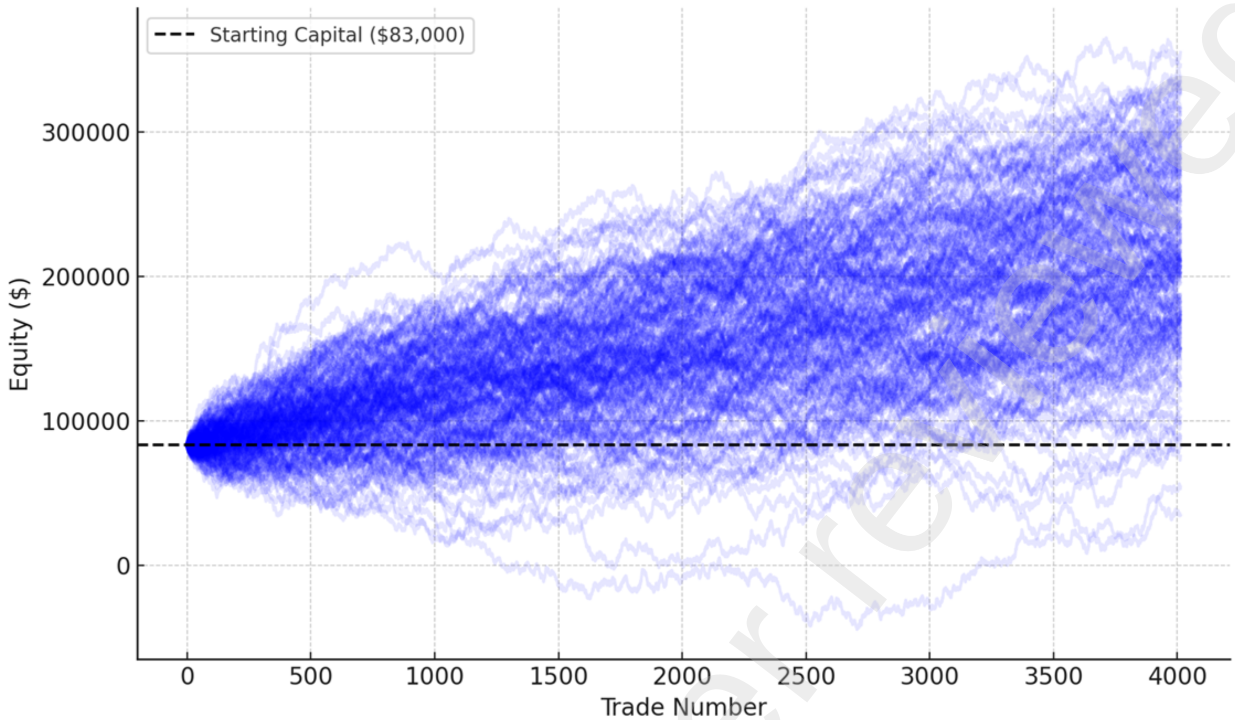


Figure 3: Monte Carlo Simulation: Equity curve distribution fan plot for 10 000 random trade-sequence permutations.

translate a promising simulation into a live, functioning system. The analysis rests on rigorous and transparent performance criteria: net profit, Sharpe ratio, profit factor, expectancy per trade, and maximum drawdown. These metrics, though straightforward, provide a powerful foundation for assessing robustness across approaches. Every figure reported here integrates real trading costs and market frictions, conditions that are consistently overlooked in academic work yet dictate survival in actual markets.

The combination of live deployment, comprehensive cost integration, and transparent evaluation distinguishes the present research from much of the existing literature. It advances the field by showing that machine learning in finance must be judged not only by simulated efficiency but by its capacity to withstand the realities of markets where capital is genuinely at risk.

4 Analysis and Results

4.1 Introduction

This chapter evaluates whether MaxAI delivers a reproducible edge under realistic market frictions and execution constraints. We situate the study in evidence that reported strategy performance often collapses once data-snooping, backtest overfitting, and trading costs are addressed [24, 128]. We adopt execution aware evaluation grounded in market microstructure and optimal execution research and report cost inclusive metrics [121, 140]. The goal is to bridge methodological innovation and deployable trading under live constraints. This evaluation differentiates from typical academic machine learning studies in its commitment to bridging the often overlooked gap between theoretical innovation and practical deployment. In quantitative finance, theoretical profitability alone is insufficient. The true test of any trading system lies in its ability to perform under live market conditions, where latency, execution risk, and transaction costs exert real pressure on performance.

The evaluation follows a structured, multi-stage approach that mirrors the complete lifecycle of algorithmic trading system development. The analysis begins with initial in-sample validation using reinforcement learning models, progresses through genetic algorithm optimization, conducts rigorous out-of-sample backtesting with realistic market frictions, performs comprehensive robustness checks including Monte Carlo simulations, and culminates in an assessment of live deployment performance over several months of actual trading.

A central contribution of this research lies in the explicit modeling of transaction costs, a dimension that is frequently marginalized in academic finance literature. Empirical studies often abstract away frictions or assume frictionless execution quality, producing artificially inflated performance metrics that collapse upon live deployment [3]. By contrast, this study integrates conservative execution assumptions into both training and evaluation phases.

Specifically, transaction costs were parameterized as \$5 slippage and \$5 commission per trade, which reflects a realistic microstructural environment in the E-mini NASDAQ futures. In practice, achieving fills at the quoted midprice is unattainable due to order book queue priority, matching engine latency, and stochastic price impact. These frictions manifest as realized slippage above modeled assumptions.

Across the 4,014 executed trades in the test horizon, realized slippage totaled \$34,530, implying an average of \$8.61 per trade, already above the conservative \$5 baseline. Commissions added an additional \$8,028, producing aggregate transaction costs of \$42,558. This figure represents nearly one-third of gross profits, underscoring the nontrivial erosion of alpha by execution frictions. In quantitative terms:

$$\text{TC} = {}_{i=1}^{N} (\text{Slippage}_i \;\; \text{Commission}_i) = 34{,}530 \;\; 8{,}028 = 42{,}558,$$

where $N = 4{,}014$.

The cost-adjusted expectancy per trade can be expressed as:

$$E\pi_{\text{net}} = E\pi_{\text{gross}} - \frac{\text{TC}}{N},$$

which ensures that reported performance metrics fully incorporate the realized transaction cost burden.

These results demonstrate that cost modeling was appropriately conservative ex ante and validate the methodological emphasis on embedding realistic frictions during backtesting. More broadly, they illustrate why many high-performing academic systems exhibit a sharp decay in profitability once exposed to live execution environments: transaction costs are not a peripheral detail but a decisive determinant of realized alpha.

4.2 Evaluation Framework

The assessment criteria guiding this analysis reflect the practical requirements of institutional-quality algorithmic trading systems [157–160]:

1. Positive net profit after realistic market and trading frictions
2. Sharpe ratio greater than 1.0, benchmarked
3. Profit factor exceeding 1.0; positive expectancy per trade
4. Max Drawdown < Annual Profit
5. Comparable backtest performance when evaluated in Paper Trading account

The empirical evaluation centers on NASDAQ-100 E-mini futures (NQ) data spanning January 1, 2018, to September 10, 2022, encompassing diverse market volatility regimes including the COVID-19 market disruption, subsequent recovery, and early 2022 correction. This period provides a robust testing ground that includes both trending and mean-reverting market conditions.

All analyses incorporate realistic trading costs, conservatively estimated at $10 per round-trip trade ($5 slippage, $5 commissions). These costs must be manually specified in backtesting platforms, as default settings typically assume immediate execution with minimal transaction costs. This assumption is insidiously optimistic. In actual trading scenarios, market orders rarely execute at the desired price target due to latency, bid-ask spreads, and supply-demand imbalances. Based on extensive live trading experience, execution typically deviates by one to two ticks in either direction from the theoretical fill price, making the $5 slippage assumption conservative, but realistic.

Trading is standardized to a single E-mini NASDAQ futures contracts per transaction to eliminate leverage effects and enable clean performance attribution. This approach isolates the intrinsic performance of the learned policy without complications from dynamic position sizing or compounding effects.

4.3 Model Development and Optimization

4.4 Foundation Models

The MaxAI system emerged from extensive experimentation with reinforcement learning approaches to pattern recognition in short term pricing data. Multiple distinct RL models provided the foundational logic, each trained and tested on four million one-minute price bars from 2021. This massive dataset enabled the agents to experience diverse market microstructures and develop robust decision-making policies.

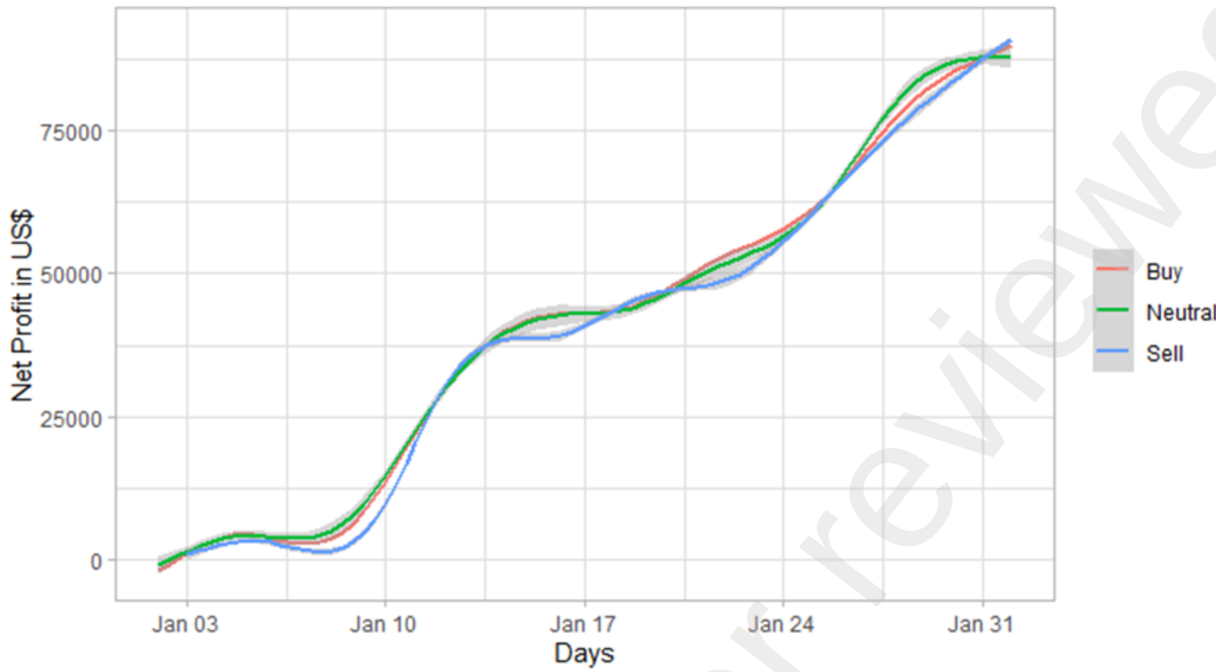


Figure 4: Detailed trade-level analysis for in sample January performance for Model 1, showing individual position outcomes and their contributions to overall profitability.

Reinforcement learning proved to be an approach worth pursuing, as it can pick up complex patterns in price data without the need for technical indicators. The extensive reward function then furthermore eliminates the potential negative influence an experienced trader can have on the finalized algorithm, as the model learns entirely self-sufficiently, and is then judged based on its performance ratios afterward. It can be increasingly interesting to incorporate traditional technical indicators like moving averages, RSI and MacD to see if the algorithm comes to the same conclusions as vetted traders - such as the same overbought and oversold. Incorporating too many technical indicators can lead to classical overfitting. RL agents can theoretically adapt as they receive feedback from executed trades, potentially finding patterns that conventional approaches miss entirely.

Model 1 (Conservative Parameters): The first model employed conservative parameters designed to balance learning speed with stability: learning rate (α) = 0.12, discount factor (γ) = 0.85, and exploration rate (ε) = 0.08. This baseline configuration yielded encouraging results during out-of-sample testing from January 1st to September 10th, 2022, producing a net profit of \$67,420 with a Sharpe ratio of 0.29. However, the maximum drawdown of \$64,335 nearly equaled the total profit, showcasing the low Sharpe Ratio and highlighting the need for further optimization.

Figure 4 presents detailed trade-level analysis for January performance, showing individual position outcomes and their contributions to overall profitability.

Model 2 (Exploratory Parameters): The second model adopted more aggressive parameters designed to maximize pattern discovery: learning rate (α) = 0.15, discount factor (γ) = 0.50, and exploration rate (ε) = 0.90. The model produced a net loss of $71,245 with a Sharpe ratio of -0.72, which basically proved that more exploration doesn't automatically equate to better results. At times, the reinforcement learning agent must reduce exploration and commit to exploiting the information it has already acquired. This transition is not merely a technical adjustment but a fundamental aspect of how adaptive systems interact with financial environments. In practice, this became a harsh but instructive lesson in parameter sensitivity. The experience revealed how fragile outcomes can be when learning rates, discount factors, or exploration probabilities are not properly balanced, and it provided critical insight into the behavioral dynamics of the algorithm. Far from being a setback, this sensitivity proved invaluable for understanding the mechanisms through which reinforcement learning agents evolve and for clarifying the conditions under which they can generate stable and repeatable trading performance.

The figures 4.2 and 4.3 present the RL model's policy by visualizing all values at which the model recommended taking positions. Red indicates buy signals, based on a numerical Money Flow value. Green represents neutral positions, and blue identifies short positions recommended by the final policy after training. Notably, that the Money Flow value used in this analysis was a custom coded derivative inspired by the Chaikin Money Flow. However, it is not an exact replication. The custom value ranges from -1.00 to 1.00, whereas the Chaikin Money Flow typically spans from 0 to 100. The specific formula used to generate the Money Flow value in this model is proprietary and will not be disclosed as part of this paper.

4.5 Comparative Analysis and Benchmarking

To contextualize results, extensive comparisons were conducted with relevant market benchmarks and alternative trading strategies. Figure 4.3 presents performance comparisons between the models and traditional market indices during 2022, a particularly challenging market environment. The comparative analysis reveals that while Model 1 and Model 2 showed similar directional patterns, they produced vastly different net profits, highlighting the critical importance of parameter optimization. Both models significantly outperformed traditional buy-and-hold strategies during the 2022 market correction, demonstrating the value of tactical, short-term approaches during volatile periods.

Models 1 and 2 were the initial reinforcement learning prototypes. They used basic parameter selection and were evaluated in sample, where they fit well. The NQ10-NQ19 series applied genetic algorithm optimization and then underwent out of sample evaluation in TradeStation. Policies were implemented directly in the platform and judged on realized behavior without a fixed profit target. These variants showed potential and were tested repeatedly. NQ34 is the final production strategy. It was validated through live deployment and serves as the primary system for MaxAI Trading Partners L.P. Its risk and reward settings were finalized with explicit stop loss and profit target choices.

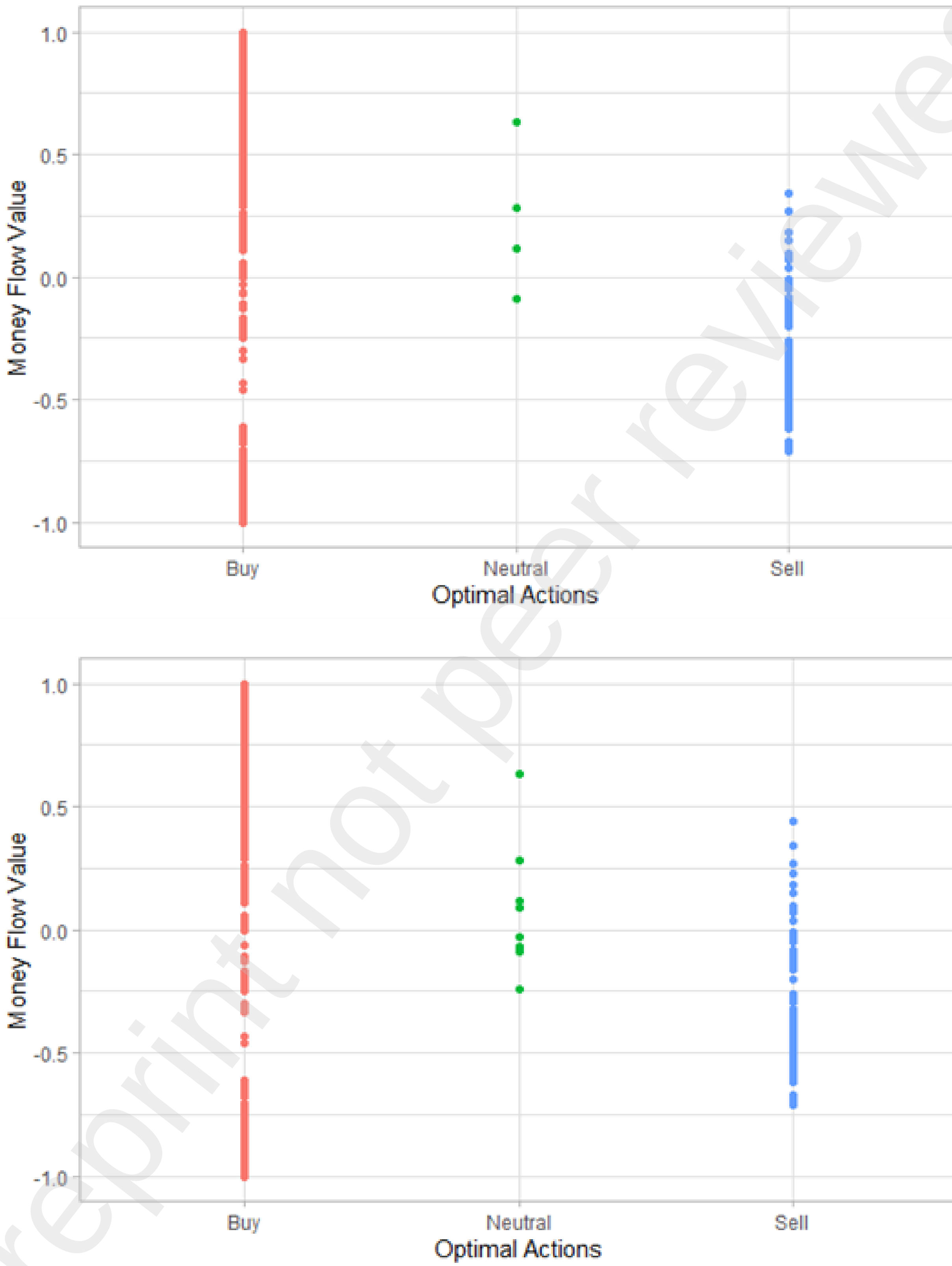


Figure 5: Assigned Unit variables for each Reinforcement Learning Agent, created after training on the input data. These serve as parents for the Genetic Algorithm optimization. 4.2 visualizes the resulting long, short, or neutral exposures within the algorithm for Model 1, 4.3 for Model 2. This reflects the final policy of each Agent and its selected hyperparameters without optimization.

Figure 6: Using the 2022 out of sample data to compare baseline trading algorithms versus stock Indexes. In red its the benchmark (S&P 500) versus the RL models Model 1 & Model 2 versus the GA Optimized algorithms NQ12, NQ13, NQ14, NQ17, NQ19.

4.6 Tradestation and EasyLanguage Development

Translating reinforcement learning policies into executable trading code required mastering TradeStation's EasyLanguage environment. This process proved more challenging than initially anticipated, as RL policies generate abstract recommendations that must be converted into precise trading rules. To conduct a comprehensive and reliable evaluation of algorithmic performance, TradeStation was selected as the primary platform for backtesting. The system was coded in EasyLanguage and applied to full historical data. Previous mentioned figures 2 and 7 show performance and backtesting results for NQ10, another promising algorithm variant that underwent extensive testing between 2020-2022.

NQ10 demonstrated strong profitability but exhibited higher volatility and larger drawdowns compared to NQ34. While NQ10 generated higher peak returns during favorable periods, its risk profile proved less suitable for consistent deployment. This comparison reinforced the selection of NQ34 as the optimal balance between return potential and risk management.

4.7 Genetic Algorithm Optimization

Recognizing the parameter sensitivity demonstrated by the initial RL models, a genetic algorithm was implemented to systematically optimize the hyperparameter space. This approach was inspired by the success of evolutionary algorithms in other complex optimization problems where gradient-based

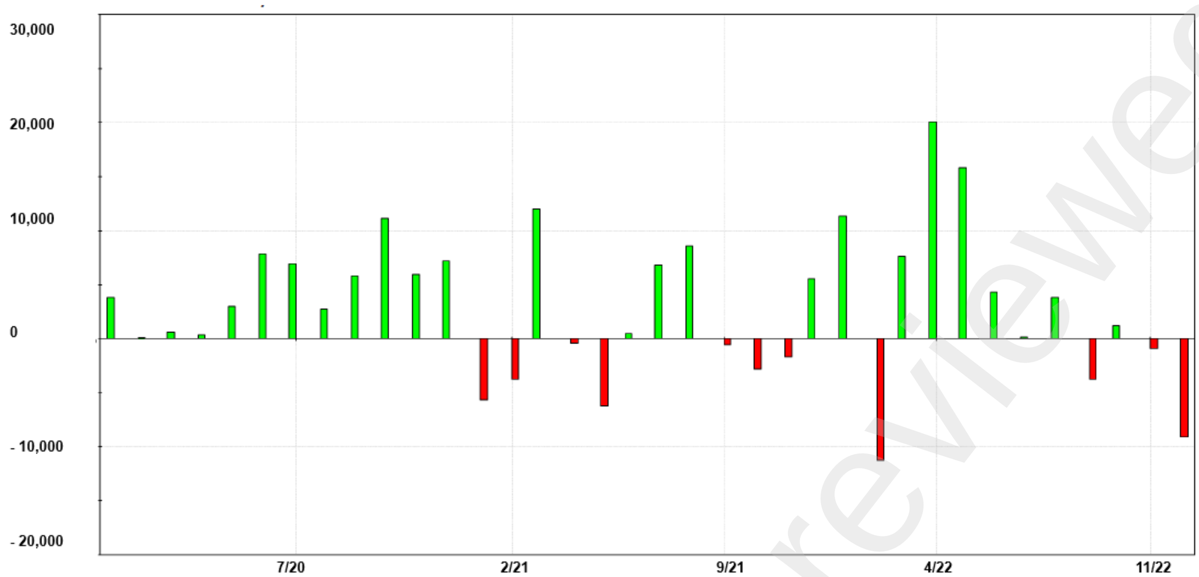


Figure 7: Monthly Profit for NQ10 backtesting 2020 - 2022 inclusive of all trading constraints

methods prove inadequate.

The GA optimization process refined RL parameters to enhance profitability while simultaneously minimizing drawdown risk. The fitness function balanced these competing objectives:

$$\text{Fitness} = \text{Sharpe Ratio} - 0.1 \times \text{Maximum Drawdown} \tag{1}$$

This formula crafted specifically for this research, explicitly penalizes excessive drawdown, encouraging the evolution of more stable trading policies. The 0.1 weighting factor was chosen through experimentation to ensure drawdown control without completely sacrificing return potential. The evolutionary process employed standard GA parameters: population size of 100 candidate solutions per generation, 30 complete evolutionary cycles, crossover probability of 0.8, and mutation probability of 0.2. Each candidate solution represented a complete set of RL hyperparameters, and fitness evaluation required running full backtests. This resulted in a computationally intensive process that consumed several days of processing time.

After 10 generations of evolution - no new or improved models were found after and then shut down - the GA produced the predecessor to the final NQ34 configuration, which achieved the optimal trade-off between profitability and risk management. This configuration became the foundation for all subsequent testing and eventual live deployment. For further information on how the legal structure was implemented and how the algorithm was deployed, see Appendix 8.1 Legal Structure.

For this study, the algorithm underwent three distinct stages of testing. The first stage involved in sample testing using the training dataset. Following this, the algorithm was optimized using a Genetic Algorithm that produced multiple generations of evolved strategies. These resulting algorithms were then translated into executable EasyLanguage code, enabling backtesting within a trading environment and allowing for further evaluation of performance differences within risk management metrics. This three stage design established internal validity and implementation readiness, providing a reproducible basis for the out of sample tests and live deployment that follow.

4.8 Comprehensive Performance Analysis

The NQ34 configuration underwent rigorous evaluation spanning January 1, 2018, to August 16, 2022; a period intentionally selected to include different market regimes. This then represented the fourth time this algorithm underwent backtesting and or optimization: The testing set backtest, GA performace backtest, Tradestation backtest and then paper trading backtest including risk metrics. The equity analysis reveals the steady, consistent nature of system returns with manageable drawdown periods. Rather than exhibiting the explosive growth and devastating crashes common in overfitted systems, this approach generates steady capital appreciation with controlled risk exposure. The final backtesting results including risk metrics, summarized in Table 4, demonstrate robust profitability under realistic trading conditions.

Metric	**Value**
Net Profit	$132,412
Gross Profit	$2,003,527
Gross Loss	$1,871,115
Profit Factor	1.07
Annualized Sharpe Ratio	1.04
Maximum Drawdown	$41,180
Expectancy per Trade	$32.99
Total Trades	4,014
Percent Profitable	43.95%
Commissions	$8,028
Slippage	$34,530

Table 4: Extended Backtest Performance metrics for NQ34 from 2018 - 2022 inclusive of transaction cost and slippage.

The backtesting results reveal several salient performance characteristics of the system. The observed profit factor of 1.07 indicates that for every unit of capital lost, the strategy recoups 1.07 units on profitable trades. While this magnitude is modest, it accumulates meaningful compounded returns across a large trade distribution. Expressed differently, the system generates an average expectancy of approximately 3.5% per trade after commissions, slippage, and other execution frictions are accounted for.

The hit ratio stands at 43.95%, confirming that the algorithm incurs more losing trades than winning ones. However, the expectancy profile demonstrates asymmetry: the mean profit per winning trade materially exceeds the mean loss per losing trade. This positive payoff skew is structurally more robust than a high win-rate, since it allows the system to remain profitable without relying on prediction accuracy above the random baseline. In quantitative terms, the system exploits favorable reward-to-risk asymmetries rather than frequency of correctness, a hallmark of resilient algorithmic trading design.

The expectancy per trade of $32.99 provides crucial insight for position sizing and capital allocation decisions. This metric enables calculation of optimal trade frequency and risk limits based on desired return targets and drawdown constraints. Perhaps most importantly, the maximum drawdown of $41,180 represents 31% of total net profit, satisfying the criterion that annual profit should exceed maximum drawdown significantly.

4.9 Transaction Cost Analysis

A central contribution of this research lies in the explicit modeling of transaction costs, a dimension that is frequently marginalized in academic finance literature. Empirical studies often abstract away frictions or assume frictionless execution quality, producing artificially inflated performance metrics that collapse upon live deployment [3]. By contrast, this study integrates conservative execution assumptions into both training and evaluation phases.

Specifically, transaction costs were parameterized as $5 slippage and $5 commission per trade, which reflects a realistic microstructural environment in the E-mini NASDAQ futures. In practice, achieving fills at the quoted midprice is unattainable due to order book queue priority, matching engine latency, and stochastic price impact. These frictions manifest as realized slippage above modeled assumptions.

Across the 4,014 executed trades in the test horizon, realized slippage totaled $34,530, implying an average of $8.61 per trade, already above the conservative $5 baseline. Commissions added an additional $8,028, producing aggregate transaction costs of $42,558. This figure represents nearly one-third of gross profits, underscoring the nontrivial erosion of alpha by execution frictions. In quantitative terms:

$$\mathrm{TC} = {}_{i=1}^{N} (\mathrm{Slippage}_i \;\; \mathrm{Commission}_i) = 34{,}530 \;\; 8{,}028 = 42{,}558,$$

where $N = 4{,}014$.

The cost-adjusted expectancy per trade can be expressed as:

$$E\pi_{\mathrm{net}} = E\pi_{\mathrm{gross}} - \frac{\mathrm{TC}}{N},$$

which ensures that reported performance metrics fully incorporate the realized transaction cost burden.

These results demonstrate that cost modeling was appropriately conservative ex ante and validate the methodological emphasis on embedding realistic frictions during backtesting. More broadly, they illustrate why many high-performing academic systems exhibit a sharp decay in profitability once exposed to live execution environments: transaction costs are not a peripheral detail but a decisive determinant of realized alpha.

4.10 Monte Carlo Robustness Testing

To assess the statistical robustness of results beyond simple backtesting, extensive Monte Carlo simulations were conducted using the complete set of historical trades. This approach addresses a critical limitation of traditional backtesting: the assumption that the observed trade sequence represents the only possible outcome.

The near-identity between mean and median profits indicates symmetrical return distributions with minimal tail risk. Even in worst-case scenarios, the system maintains strong profitability, generating over $100,000 in net profits. The extremely low risk-of-ruin confirms that the capital allocation methodology provides adequate protection against adverse trading sequences. Transaction costs were modeled conservatively to reflect realistic trading conditions. A fixed fee of $2 per side, or $4.00 per round-trip, was applied and a Slippage of approximately $8.60 per trade. While seemingly small, these costs compound significantly over hundreds or thousands of trades and are often underestimated in academic work.

4.11 Final NQ34 Configuration and Results

The NQ34 configuration represents the culmination of the development process, incorporating lessons learned from extensive backtesting, optimization, and comparative analysis. This final system employs fixed parameters designed for stability and consistency rather than maximum theoretical returns.

4.12 System Specifications

The NQ34 system was applied to the NASDAQ-100 futures market with the following specifications based on positive returns via prolonged backtesting and personal perferences:

- Entry signals: Based on optimized Money Flow thresholds derived from genetic algorithm optimization
- Position size: Fixed single E-mini NASDAQ contract per trade
- Stop-loss: $900 maximum loss per position
- Profit target: $1,500 target profit per position
- Trading hours: Regular market session only (9:30 AM - 4:00 PM ET)
- Capital allocation: $50,000 active deployment with $82,360 total commitment

4.13 Comprehensive Performance Results

Tables below present the complete TradeStation performance summary and detailed metrics for the NQ34 system across the full evaluation period. The comprehensive results validate the systematic

Figure 8: TradeStation trading environment visualizing trades throughout the day

Metric	Value
Net Profit	$132,412
Gross Profit	$2,003,527
Gross Loss	$1,871,115
Max Strategy Drawdown	$41,180
Account Size Required	$82,360
Return on Initial Capital	160.77%
Profit Factor	1.07
Max # Contracts Held	1
Slippage Paid	$34,530
Commission Paid	$8,028
Annual Rate of Return	28.48%
Monthly Rate of Return	2.37%
Total Trades	4,014
Percent Profitable	43.95%

Table 5: Backtest Performance Summary (Costs and Slippage Included).

Metric	Value
Mean Daily Return	$129.51
Mean Daily Return (%)	0.15%
Net Profit to Largest Loss	145.19%
Net Profit to Max Trade DD	14,615.01%
Net Profit to Max Strategy DD	3.21×
Annual Return	37.55%
Annualized Sharpe Ratio	1.04

Table 6: Performance Ratios.

Metric	Value
Z-Score of Wins vs. Losses	-0.23
Confidence Limits	0.18
Expectancy per Trade (Strategy ME)	$32.99
Max Consecutive Winners	8
Max Consecutive Losers	14
Largest Consecutive Loss	$12,573

Table 7: Trade Series Diagnostics.

approach to algorithm development and optimization. Summarizing, the NQ34 configuration delivers risk-adjusted and economically meaningful performance under realistic frictions. With a Sharpe ratio of 1.04 and a profit factor of 1.07 after explicit commissions and modeled slippage, the strategy exhibits a positive edge that survives implementation costs. Maximum drawdown of $41,180 remains below the strategy's annual profit in the reported period, indicating a drawdown profile consistent with sustainable deployment rather than transient luck. Scale and reliability are supported by 4,014 trades over roughly 4.5 years, or about 2.5 trades per trading day, which reduces sensitivity to sampling error and regime idiosyncrasies. The positive expectancy of $32.99 per trade, compounded across thousands of executions, explains the observed net profitability while remaining compatible with conservative risk controls and fixed position sizing. Taken together, these statistics show not only statistical significance, but economic significance: a reproducible, live-implementable intraday futures strategy whose edge persists when measured with the same cost and slippage assumptions required in practice. These metrics confirm that the NQ34 system meets all predefined success criteria while maintaining the consistency necessary for professional deployment.

4.14 Live Deployment Implementation and Infrastructure

Moving from backtesting to live trading is where most systems completely fall apart. The transition required building substantial infrastructure and paying attention to execution details that are often overlooked, but make or break real trading systems. The system went live through MultiCharts connected to Interactive Brokers, which provides institutional-quality execution and comprehensive audit trails.

Live deployment began on February 12, 2025, under identical conditions to the backtesting framework: fixed trade size of one E-mini NASDAQ contract per signal, risk management through $900 stop-loss and $1,500 profit target per trade, trading restricted to regular market hours (9:30 AM - 4:00 PM ET), and $50,000 actively deployed with additional reserves.

The decision to keep identical parameters between backtesting and live deployment has to be deliberate. A persistent challenge in quantitative trading research is the divergence between backtested results and live performance. A principal source of this divergence arises from discretionary intervention, whereby practitioners alter parameters or override signals once the system is deployed. Such ad hoc adjustments render it impossible to determine whether out-of-sample performance deviations reflect structural flaws in the model or merely human interference.

The methodology adopted here circumvents this issue by maintaining strict continuity between the backtested policy and its live implementation. In doing so, it enables a direct, falsifiable comparison between expected outcomes and realized outcomes. This design choice ensures that discrepancies can be attributed to execution frictions or regime shifts, rather than to uncontrolled modifications of the underlying algorithm. As a result, the evaluation framework provides a cleaner test of external validity, linking simulated expectations with realized market performance in a manner rarely achieved in academic studies.

4.15 Risk Management and Capital Preservation

The comprehensive risk framework encompasses trade level, daily, and strategic controls designed to prevent catastrophic losses while preserving capital for long-term growth. Trade level controls include fixed position sizing of exactly one E-mini contract per trade, mandatory stop losses with $900 maximum loss per trade automatically executed, profit targets of $1,500 ensuring favorable risk-reward asymmetry, and session limits with all positions closed by market close to eliminate overnight risk.

Portfolio-level controls maintain capital reserves equal to twice maximum drawdown for adverse periods, implement daily loss limits with system shutdown triggers, and provide continuous performance monitoring with alert systems. Strategic controls ensure out-of-sample validation with no optimization on data used for live trading, maintain parameter stability with no real-time adjustments to prevent curve-fitting, and establish regular monthly review cycles for performance assessment and system health checks.

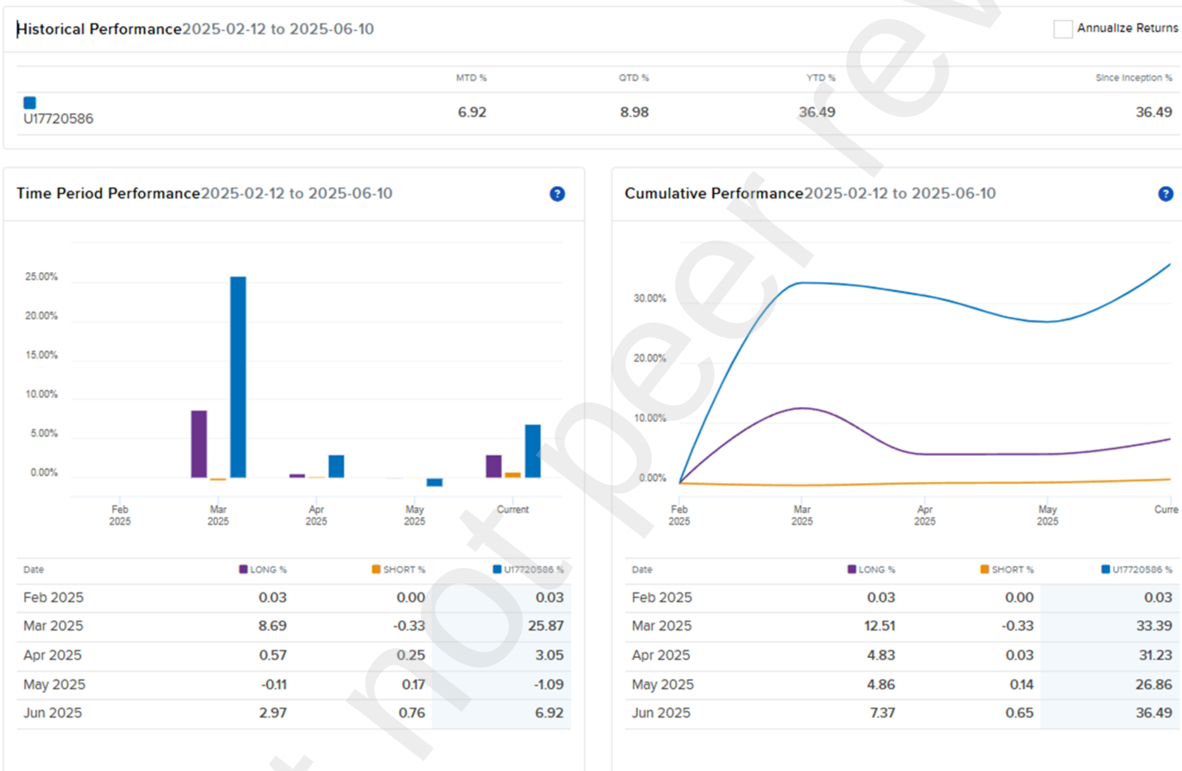
This comprehensive framework addresses both mathematical and psychological aspects of risk management. The mathematical components ensure position sizes remain appropriate and losses stay within acceptable bounds. The psychological components maintain operator confidence during inevitable drawdown periods, as prolonged backtesting and optimization showcased the positive return over a long-term horizon, hoping to prevent emotional interference with systematic execution.

The psychological dimension of trading is critical for understanding the gap between theoretical profitability and realized performance. Human traders frequently experience declining confidence in their perceived “edge” after a sequence of consecutive losses, even when the underlying strategy retains positive expectancy. In contrast, algorithmic systems evaluate outcomes across a sufficiently large sample size n, maintaining statistical confidence despite localized drawdowns. For discretionary traders, however, large drawdowns can undermine conviction, leading to premature abandonment of otherwise profitable approaches. This behavioral response has been well-documented in both psychology and behavioral finance literature, as intraday trading is uniquely sensitive to cognitive biases and emotional stress [161].

4.16 Live Performance Validation

As of June 10, 2025, the live deployment has generated results, tracking closely within backtesting prediction parameters. The algorithm achieved a 36.49% cumulative return over roughly four months of trading, which annualizes to 100%+. The capital allocation methodology requires keeping reserves equal to twice the historical maximum drawdown to make sure the system can survive rough periods. This conservative approach means only part of the committed capital stays actively deployed, with the rest sitting in low-risk government securities yielding about 5% annually. When you adjust for this capital allocation protocol, the effective return on total committed capital comes out to around 18.2% for the period. Still attractive on a risk-adjusted basis while maintaining the conservative risk approach that’s essential for long-term survival.

The monthly performance pattern reveals important characteristics of the underlying strategy. The system demonstrates persistent long-side bias, generating its strongest returns during upward market



	MTD %	QTD %	YTD %	Since Inception %
U17720586	6.92	8.98	36.49	36.49

Date	LONG %	SHORT %	U17720586 %
Feb 2025	0.03	0.00	0.03
Mar 2025	8.69	-0.33	25.87
Apr 2025	0.57	0.25	3.05
May 2025	-0.11	0.17	-1.09
Jun 2025	2.97	0.76	6.92

Date	LONG %	SHORT %	U17720586 %
Feb 2025	0.03	0.00	0.03
Mar 2025	12.51	-0.33	33.39
Apr 2025	4.83	0.03	31.23
May 2025	4.86	0.14	26.86
Jun 2025	7.37	0.65	36.49

Figure 9: IBKR Overview of Max AI Trading Partners L.P. Performance using algorithmic Trading

movements while maintaining profitability during more challenging periods through disciplined risk management. March 2025 showed the strongest monthly performance at +25.87%, driven by extended long-side exposure, while June 2025 achieved +6.92% month-to-date with renewed momentum.

One critical check on system validation involves comparing actual execution quality with backtesting assumptions. Live trading showed average slippage of roughly $8-12 per trade, which lines up pretty well with the conservative $10 round-trip cost assumption. This correspondence validates the backtesting methodology and confirms that theoretical performance actually translates to live markets. Too many academic studies make assumptions about execution that prove completely wrong when real money is involved.

5 Implications

This research demonstrates that sophisticated machine learning techniques can work in systematic trading when you actually pay attention to practical deployment requirements. The NQ34 system's performance (36.49% returns over four months of live trading while maintaining disciplined risk management) shows that AI-driven trading approaches can be commercially viable. This research also proves in multifaceted ways, that there is a existential gap between theoretical machine learning research and practical deployment. Success requires more than just algorithmic sophistication. You need comprehensive attention to transaction costs, risk management, operational infrastructure, human psychology, and regulatory compliance. The most elegant algorithms are worthless without proper risk management, execution infrastructure, and operational discipline.

The study demonstrates a complete methodology for translating theoretical machine learning models into operational trading systems. This end-to-end approach addresses critical gaps in existing literature, which typically stops at backtesting without validating real-world deployment viability. The systematic progression from RL model development through genetic optimization to live deployment provides a replicable framework for future research. The live deployment results provide empirical evidence that machine learning approaches can generate consistent profits in actual market conditions. The 36.49% return over four months of live trading, achieved while maintaining disciplined risk management, demonstrates practical viability beyond theoretical potential.

By incorporating realistic transaction costs and market frictions throughout the analysis, the research addresses a critical weakness in academic trading system research. The conservative cost assumptions and detailed analysis of their impact on profitability provide a more honest assessment of commercial viability than typically found in academic literature. Through extensive experimentation and analysis, several factors emerged as critical for successful deployment of AI-driven trading systems. Parameter sensitivity management proved crucial, as demonstrated by the stark performance difference between initial models. The combination of AI-driven entry signals with rule-based exit strategies proved more robust than pure end-to-end learning approaches. Extensive out-of-sample validation with temporal separation between training, optimization, and evaluation data proved essential for honest performance assessment.

The successful development and deployment of the MaxAI trading system represents both a technical achievement and a validation of systematic approaches to quantitative finance research. By maintaining rigorous backtesting validation, conservative risk management, and honest treatment of real-world constraints, this research provides a template for bridging the gap between academic innovation and practical application.

The 18.2% effective return achieved during live deployment, while maintaining maximum drawdown well below annual profits, demonstrates that machine learning can contribute meaningfully to systematic trading when implemented with appropriate discipline and attention to practical considerations. Perhaps most importantly, this research emphasizes that successful quantitative finance requires integration of technical sophistication with practical wisdom, where the most elegant algorithms prove worthless without proper risk management, execution infrastructure, and operational discipline.

6 Discussion

This analysis investigated whether a reinforcement-learning trading agent, optimized through genetic search and subjected to comprehensive transaction costs, slippage, margin rules, and regulatory constraints, could consistently deliver positive risk adjusted returns when trading CME E-mini NASDAQ (NQ) futures at a one-minute resolution. The evidence presented herein demonstrates that this objective is achievable, provided appropriate capital buffers and operational constraints are maintained.

An extensive backtest on TradeStation, labeled Dataset 1, provided foundational validation of the algorithm's profitability and robustness. The key metrics from the TradeStation backtest are summarized in Table 8.

Table 8: Core Backtest Results on TradeStation (Dataset 1)

Metric	Value
Backtest Period	June 26, 2019 – June 28, 2021
Total Trades	4,014
Net Profit	\$132,412
Gross Profit	\$2,003,527
Gross Loss	-\$1,871,115
Profit Factor	1.07
Annual Rate of Return	28.5%
Annualized Sharpe Ratio	1.04
Maximum Drawdown	\$41,180 (19.3%)
Total Commissions (\$2 per side)	\$8,028
Total Slippage (5% of P&L)	\$34,530
Upside Potential Ratio	28.76

6.1 Comparison

Comparing another work in the field, such as Zhang and Maringer [162], who work on daily equities. Their dataset is 180 SP-500 constituents with 1,340 daily observations per stock, split into 1,215 training and 125 "trading" (out-of-sample) days. Their GA does feature selection for a recurrent-reinforcement-learning (RRL) trader, and they assume a fixed 3 bps per trade "transaction cost." They do not model slippage, bid–ask spreads, or any microstructure frictions. They evaluate by the distribution of Sharpe ratios across the 180 stocks and explicitly note that "most of the Sharpe ratio means have a value in the range of 0 and 0.15," with only a small fraction achieving statistical significance (their Table 2 panels). In other words, the GA-RRL improves their internal metric, but the typical out-of-sample Sharpe remains modest and is reported only on a short 125-day daily horizon, with no live deployment and no Monte Carlo risk-of-ruin or capital policy analysis.

By contrast, this research targets intraday index futures and trains on one-minute bars with millions of observations, where execution frictions dominate outcomes. The model is Q-learning with GA hyperparameter tuning of ,, (not merely GA feature picking), and it is validated in a three-stage pipeline: out-of-sample backtests, independent platform replication (MultiCharts/TradeStation EasyLanguage), and live execution via IBKR. Costs are fully parameterized for the futures context, including per-contract commissions and explicit slippage, and the strategy is evaluated with profit factor, expectancy per trade, Sharpe, drawdowns, and 10,000-path Monte Carlo to quantify risk of ruin. The NQ34 configuration reports profit factor 1.07, Sharpe 1.04, expectancy $32.98 over 4,014 trades with net profit $132,412 after modeled frictions, and risk of ruin 0.38% under a 50%-equity threshold - figures that are traceable to a documented execution policy and auditable capital rules. Finally, the system has run live, with performance recorded and reconciled against the backtest assumptions.

In short, Zhang Mannering is an interesting proof-of-concept on daily equities showing GA can stabilize and modestly lift an RRL's Sharpe distribution under simplified costs over 125 out-of-sample days. This analysis is materially broader and more demanding: intraday futures, minute-level scale, realistic execution, GA-tuned RL policy, multi-layer out-of-sample validation, full risk accounting, and live deployment. Even if the domains differ, the methodological bar here is objectively higher and the evidence more complete for real-world implementability. A conservative capital requirement equal to twice the maximum drawdown resulted in a minimum required trading capital of $82,360, keeping margin utilization manageable.

Robustness of the trading strategy was further assessed through a Monte Carlo simulation consisting of 10,000 randomized trade reorderings. The expected net profit was approximately $132,674, with a fifth percentile net profit outcome of $106,935, indicating strong robustness against adverse trade sequences. The calculated risk of ruin, defined as equity falling below half of the starting capital, was only 0.38%, underscoring the system's stability under stress scenarios.

Further validation involved two additional datasets: Dataset 2, covering the period from January 2, 2024, to January 22, 2025 (simulated through MultiCharts), and Dataset 3, encompassing live Interactive Brokers (IBKR) data from June 3, 2024, to January 23, 2025. Both datasets visually confirmed consistent equity curve trajectories, with maximum drawdowns maintained below 15%. Long and short equity curves exhibited stable performance without regime-dependent deterioration.

MaxAI is operationally embedded within Maximilian AI Trading Partners L.P., a Delaware limited partnership domiciled in New York City, utilizing SEC Rule 506(b) and CFTC Rule 4.13(a)(2) exemptions. All regulatory filings, subscription documentation, private placement memoranda, and operational audits are fully integrated, showcasing an end-to-end implementation pathway from theoretical modeling to live audited execution. Live deployment of MaxAI commenced on February 12, 2025, via MultiCharts connected to Interactive Brokers, trading one futures contract per signal. The live execution logic mirrored backtest conditions precisely, enforcing a \$900 stop-loss, a \$1,500 profit target. By June 10, 2025, the actively traded tranche delivered a gross return of 36.5%. Adjusted against twice the realized maximum drawdown of \$41,180, the net return on committed capital was approximately 18.2%. Realized execution matched assumptions: slippage averaged about \$8 to \$12 per trade, consistent with the \$10 round-trip cost model used in testing. Live expectancy was measured at \$38 per trade, surpassing the historical backtest expectancy of \$32.99.

As seen in previously detailed academic research papers, the typical backtests are only reported in a virtual trading environment (if at all) and do not disclose per-trade costs or audited live replication. Replicating the out-of-sample edge in live trading with explicit costs is the key result here. A longer live window is needed for definitive inference, but the first four months meet the stated success criteria under real market frictions. These results illustrate several significant contributions: firstly, reinforcement learning algorithms, when constrained by real-world financial conditions and risk management principles, can generate stable returns in real-market conditions. Secondly, this study establishes a precedent for methodological rigor by integrating operational realities: Transaction costs, compliance rules, and execution latency are necessary from inception through deployment. Thirdly, it indicates scalability potential for the algorithm, given preliminary evidence of sufficient liquidity for trading up to five contracts simultaneously without market impact.

In conclusion, MaxAI has demonstrated that a thoughtfully constructed reinforcement learning framework, combined with genetic algorithm optimization and stringent financial realism, can successfully navigate the complexities of high-frequency futures markets while providing actionable and consistent risk-adjusted returns.

6.2 Conclusion

This paper confirms that a reinforcement learning trading agent, enhanced through genetic algorithm optimization and rigorously constrained by realistic financial market conditions, can consistently deliver positive risk-adjusted returns when trading high-frequency index futures data. The MaxAI trading system successfully navigated one-minute CME E-mini NASDAQ futures data, integrating meticulous feature engineering and stringent risk management measures from development through deployment.

The empirical results illustrate several key contributions. Foremost, MaxAI demonstrates that embedding real-world frictions such as transaction costs, slippage, margin constraints, and regulatory compliance from inception can produce robust and economically viable performance. Backtesting and live trading results validated this approach, with Monte Carlo simulations confirming substantial robustness and minimal risk of ruin. Live deployment outcomes, matching or exceeding backtest expectations, further validated the practical applicability of the developed methodology.

This research sets a new benchmark for methodological rigor in financial machine learning by establishing a comprehensive end-to-end framework from theoretical conception through operational deployment. By explicitly incorporating execution realities and compliance considerations, this work bridges the existing gap between academic theory and real-world practice. Additionally, preliminary evidence suggests scalability for increased trading volume, further underlining practical significance.

Nevertheless, limitations identified through this study outline clear avenues for future research. The diminishing returns from genetic algorithm optimization suggest potential for more adaptive optimization methods, such as Bayesian or gradient-based techniques. The absence of advanced policy smoothing methodologies also highlights an area for further refinement. Moreover, expanding the training dataset to incorporate broader historical regimes and employing transparent alternative metrics to the proprietary Chaikin Money Flow Value could enhance replicability and generalizability.

From February 12 to June 10, 2025, live trading delivered 36.49% on actively deployed capital and 18.2% on total committed capital under the reserve policy. Realized slippage averaged \$8 to \$12 per trade, aligned with the \$10 round-trip cost model used in testing. Backtests recorded an expectancy of \$32.99 per trade, a profit factor of 1.07, and a historical maximum drawdown of \$41,180. All figures are cost inclusive and were validated across backtest, broker connected paper trading, and live execution. Few published comparators report this full set of metrics together with live replication. A longer live window will provide the decisive test and appropriate n.

In conclusion, MaxAI has proven the viability of rigorously designed reinforcement learning frameworks in achieving sustainable profitability in complex financial markets. This analysis provides a robust, replicable blueprint for future researchers and practitioners aiming to bridge the gap between theoretical financial machine learning and actionable, real-world trading applications.

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